

Strategic (Inconsistent) Disclosures and Sophisticated Investors: Evidence from Hedge Funds

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ABSTRACT

Recent SEC regulations require that qualified hedge fund advisers provide their investors with narrative disclosures of their business and operations. We find that 40% of these disclosures omit or de-emphasize information regarding advisers' operational and investment risks when compared to other sources of public information. Funds with such "inconsistencies" are associated with predictably lower fund performance but do not differ in their fund flows, flow-performance relation, ownership structure, or management

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fees. These results are consistent with investors being subject to limited strategic thinking, which prevents them from fully unraveling the implications of strategic omissions. This, in turn, contributes to advisers' successful use of discretion to de-emphasize information with adverse performance implications. Our findings suggest that information processing frictions can facilitate nondisclosure, even in markets with sophisticated investors.

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1. Introduction

Markets dominated by sophisticated investors are significantly less regulated than public equity markets. A prevailing view in support of this limited regulatory oversight is that sophisticated investors have a solid understanding of the risks involved and do not need as much protection (see discussion by Zingales [2009]).¹ The hedge fund industry represents one such market and remained largely unregulated until the Great Recession of 2008, even as its assets under management (AUM) grew to approximately 20% of U.S. GDP (see figure 1). While limited oversight may be justified for markets with informed investors, this regulatory choice raises an important question: How effectively do sophisticated investors monitor disclosures in such markets? This paper investigates whether hedge funds exercise discretion in their public disclosures and the extent to which their investors are able to unravel the implications of such discretion.

We answer these questions using a recent change in the disclosure regulation for hedge funds. In an effort to make the hedge fund industry more transparent, the Dodd-Frank Act of 2010 mandated that hedge fund advisers with more than \$100 million in AUM register with the SEC. These funds must now provide current and potential investors with, among other things, a narrative discussion of their advisory business as well as material operational and investment risks in Form ADV Part 2 (also known as “the investment adviser brochure” or simply “brochure”). Advisers have discretion over what constitutes material information, and the SEC does not review these brochures. Prior research finds evidence consistent with hedge funds manipulating returns (Bollen and Pool [2009], Honigsberg [2019])

¹ According to SEC Rule 506, “an offering is exempt from securities regulation, if it is limited to accredited investors and no more than 35 nonaccredited investors. Furthermore, each nonaccredited investor in the offering has to be sophisticated, that is, have such knowledge and experience in financial and business matters that he is capable of evaluating the merits and risks of the prospective investment, either in his own right or with the aid of one or more purchaser representatives” (Zingales [2009], p. 404). Congress' official justification for this rule is the belief that the “public benefits are too remote” (H.R. Rep. No. 85, 73d Cong., 1st Sess. 5, 7, 15-16 (1933)). Zingales [2009] concludes that private markets, including hedge funds, are under-regulated, leading to a general lack of transparency (pp. 411–413).

and providing inaccurate information to due diligence firms (Brown et al. [2012b]). We thus expect that advisers also have incentives to strategically omit negative information from their brochures to attract and/or retain investors' capital. However, as discussed below, the extent to which this simple strategy can succeed is unclear.

On one hand, strategic omissions of negative information from brochures might not be a viable equilibrium strategy. Advisers must also disclose both material and immaterial information on key operational risk metrics in Form ADV Part 1, which takes a check-the-box and fill-in-the-blank format and is reviewed by the SEC, and fund investment risk can be partially observed through historical fund returns. Advisers that omit information from their brochures can thus be flagged by comparing their brochures with these public information sources. Hedge fund investors who observe these omissions can potentially respond appropriately to their implications (Grossman [1981], Milgrom [1981]). As such, strategic omissions would *not* help advisers attract or maintain capital, which would disincentivize advisers from engaging in such behavior.

On the other hand, strategic omissions of negative information might be a viable strategy even if investors observe them. Research on cursed equilibrium and level-K thinking suggests that even sophisticated economic agents are subject to limited strategic thinking and underestimate the correlation between other players' actions and their information, leading them to underreact to other players' actions (Eyster and Rabin [2005], Crawford, Costa-Gomes, and Iriberri [2013]). This is because "in many games the required reasoning is too complex for a thinking justification to be behaviorally plausible" (Crawford, Costa-Gomes, and Iriberri [2013]).² In our context, although hedge fund investors are sophisticated, they lack access to internal hedge fund information and therefore may not be able to reliably infer the extent to which omissions are material. If hedge fund investors are subject to limited strategic thinking and underreact to the implications of strategic omissions, then strategic omissions can help maintain fund flows.

To examine hedge funds' use of discretion in their public disclosure, we identify specific items in Form ADV Part 2 where advisers have incentives to omit negative information. Specifically, prior research shows that the presence of conflicts of interest (COIs) and disciplinary histories increases operational risk and has adverse effects on fund performance (Brown et al.

² Consistent with a cursed equilibrium, Brown, Camerer, and Lovo [2012a] find that moviegoers do not fully understand that the absence of a movie rating implies that the quality of the movie is bad. Jin, Luca, and Martin [2021] find that participants in lab experiments do not fully realize that a nondisclosure of news implies bad news. Eyster and Rabin [2005] conclude that "cursed equilibrium provides a simple, tractable, and portable psychologically motivated alternative to Bayesian Nash equilibrium that can improve empirical analysis in a broad array of settings." Crawford, Costa-Gomes, and Iriberri [2013] review the literature on level-K thinking and conclude that level-K models "share the generality and much of the tractability of equilibrium analysis, but can in many settings systematically out-predict equilibrium."

[2008, 2012b]). Excessive investment risk also increases the likelihood of deteriorating fund performance. Advisers have incentives to strategically omit information regarding these topics if such omissions help maintain or improve fund flows.

Our sample contains the intersection of SEC-registered hedge fund advisers and the Lipper Hedge Fund Database (TASS), a primary source of information on fund performance and characteristics. We identify information that is either omitted from or contradicted in Form ADV Part 2 (“inconsistencies”) by comparing advisers’ brochures to their Form ADV Part 1 (which contains disclosures of both material and immaterial COIs/disciplinary histories) and their historical returns (i.e., a measure of fund risk).

Analyzing hedge fund disclosures for the years 2014–2018, we find that 38%–41% of all advisers and 60%–64% of advisers with a disciplinary history, uncommon COI, or historically volatile returns have at least one inconsistency (see section 3 for a detailed discussion of how inconsistencies are measured). To examine whether inconsistencies *on average* reflect omissions of information with negative performance implications, we examine their associations with fund performance. Lower performance for inconsistent funds would indicate that negative information is being strategically omitted or de-emphasized.

We find that funds of advisers with an inconsistency (*inconsistent funds*) realize annualized net-of-fee returns that are 3.1–6.7 percentage points lower than both consistent funds (i.e., funds of advisers that discuss disciplinary histories, uncommon COIs, or historically volatile returns) and non-problem funds (funds without the aforementioned problems). The finding is robust to multiple measures of fund performance, fund-style-times-period fixed effects, fund-level controls, and fund fixed effects. The lower performance of inconsistent funds suggests that, on average, they have more underlying problems (e.g., operational or governance issues) that affect the fund’s ongoing performance relative to advisers who are consistent with their disclosures.³ Our evidence is consistent with the notion that inconsistencies reflect advisers’ use of discretion to de-emphasize ongoing fund problems.

Next, we investigate the underlying economic frictions that enable adviser discretion. If investors are subject to limited strategic thinking and wrongly assume that omissions reflect immaterial information, fund flows for inconsistent funds would be similar to those of consistent funds. Consistent with this notion, we find no evidence that quarterly flows or flow-performance sensitivity differs for inconsistent funds. Furthermore, we find that inconsistent funds are *more* likely to use leverage, indicating increased

³ Note, we do not claim that disclosure inconsistency *causes* material fund or adviser problems or lower performance per se. Rather, we are interested in understanding whether disclosure inconsistency indicates the existence of problems that manifest in lower fund performance.

access to credit markets, and do not differ in their use of a Big 4 accounting firm or sophisticated clients; the number of direct, indirect, or controlling owners; and ownership concentration.⁴ These findings suggest that a broad spectrum of sophisticated financial market participants do not respond to inconsistencies despite their association with lower performance. Failure to correctly respond to inconsistencies, in turn, allows managers to de-emphasize fund problems and avoid sacrificing fund flows.

A second mechanism that enables adviser discretion is low litigation against funds providing inconsistent disclosures, which reduces the cost of selectively omitting information. The SEC allocates limited resources to hedge fund supervision and does not conduct formal reviews of brochures. Moreover, hedge funds tend to rely on legal disclaimers and private communications, which could reduce the likelihood of successful enforcement actions against inconsistent funds. Finally, proving that an inconsistency represents noncompliance is difficult, as it requires access to a fund's internal information, which often remains undisclosed in regulatory filings. Consistent with low litigation, we find that only 3.6% of SEC administrative proceedings and litigation releases related to advisers are attributed to or associated with adviser brochures.

Overall, our findings support the notion that hedge fund investors exhibit limited strategic thinking, preventing them from fully responding to the implications of strategic omissions of negative information. This, in turn, enables hedge funds to exercise discretion by selectively omitting adverse information. While alternative explanations may exist for either the negative association between inconsistencies and fund performance or the lack of investor response, limited strategic thinking is the most plausible explanation for both phenomena. In addition, we find no evidence that sloppiness, overconfidence, or underlying fund problems already reflected in past performance drive the negative association between inconsistencies and fund performance. Similarly, we find no evidence that the lack of investor response to inconsistencies is driven by trading frictions that hinder timely capital withdrawals.

Our research makes several contributions. First, our paper contributes to the voluntary disclosure literature. A fundamental result of this literature is that the capital market penalty associated with withholding information propels firms to disclose voluntarily, which is known as "the unraveling result" (Grossman [1981], Milgrom [1981]). A key economic force leading to the unraveling result is that capital market participants correctly understand and respond to the implications of strategic omissions. However, research on cursed equilibrium and level-K thinking suggests that economic agents in various settings often underestimate the correlation between

⁴ Although we find no difference in management or incentive fees, we cannot rule out that these sophisticated market participants could be compensated in some other way for investing in inconsistent funds.

other agents' information and their strategic actions, leading to underreaction (e.g., Eyster and Rabin [2005], Crawford, Costa-Gomes, and Iriberri [2013], Jin, Luca, and Martin [2021], Bourveau, Breuer, and Stoumbos [2024]). We build on this literature and investigate whether hedge fund investors are subject to limited strategic thinking. Our results show that hedge fund investors do not appear to recognize and react appropriately to the implications of inconsistent disclosures, even though inconsistencies are publicly observable and correlated with worse performance. Our evidence thus suggests that limited strategic thinking can prevent unraveling.⁵

We also contribute to the literature on fund advisers' strategic disclosure decisions. Prior research documents hedge funds' use of disclosure discretion in portfolio disclosures (Agarwal et al. [2013b], Aragon, Hertz, and Shi [2013], Cici, Kempf, and Puetz [2016], Shi [2017]), reported returns (Bollen and Pool [2008, 2009], Agarwal, Daniel, and Naik [2011], Cassar and Gerakos [2011], Patton, Ramadorai, and Streatfield [2015]), and the role of regulation in reducing such misreporting (Dimmock and Gerken [2015], Honigsberg [2019]). Our paper finds that fund advisers exercise discretion to de-emphasize operational and investment risks in their regulatory disclosures, and that sophisticated investors do not respond to this use of discretion even though the omissions are publicly observable.

Finally, we contribute to the literature studying hedge funds' operational risks. Brown et al. [2008] use fund characteristics from Form ADV Part 1 to construct a measure of operational risk (ω -score) and find that operational risk is associated with lower performance. Brown et al. [2012b] develop an alternative measure of operational risk combining various fund characteristics from due diligence questionnaires (e.g., audit firms, pricing, misstatements, signature controls, internal accounting). We study hedge funds' decision to be inconsistent and find that funds that provide inconsistent disclosures about disciplinary histories or COIs underperform, whereas those with disciplinary histories or COIs that fully disclose this information to investors in Part 2 do *not* underperform. Thus, it is not the presence of a fund characteristic (e.g., disciplinary history or COI) per se, but rather the decision not to disclose this information, that is predictive of poor performance.

⁵ Several studies find that limited investor attention also affects voluntary disclosure decisions and can prevent investors from correctly responding to the implications of strategic actions (Hirshleifer and Teoh [2003], Hirshleifer, Lim, and Teoh [2009], Abramova, Core, and Sutherland [2020], Zhou and Zhou [2020], Bertomeu, Hu, and Liu [2023]). However, because hedge fund investors are generally considered sophisticated and the omissions we study are publicly observable, limited attention is unlikely the primary reason explaining why investors do not appropriately respond to the implications of inconsistencies.



FIG. 1.—Hedge fund industry assets under management (AUM). This figure plots the growth of AUM for the hedge fund industry (including funds of hedge funds) from 2000 to 2019. The solid line plots AUM, and the dashed line plots AUM as a percentage of U.S. GDP. AUM data come from BarclayHedge (www.barcleyhedge.com/solutions/assets-under-management/hedge-fund-assets-under-management/), and U.S. GDP data come from the St. Louis Fed (fred.stlouisfed.org/series/GDP).

2. Institutional Background and Data

2.1 HEDGE FUND INDUSTRY AND REGULATION

The hedge fund industry has grown significantly over the past 20 years. Figure 1 plots the historical growth in AUM by the hedge fund industry from 2000–2019, with an increase from \$0.36 trillion to \$3.48 trillion, or approximately 16% of US GDP by 2019.

Despite the significant size and growth of the industry, hedge funds historically have been unregulated and required to make few if any public disclosures. Figure 2 presents a timeline of recent hedge fund regulations in the United States (see Kaal and Oesterle [2016] for a detailed discussion of the history of hedge fund regulation in the United States). The fall of Long-Term Capital Management (LTCM) in 1998 increased awareness that hedge funds pose systemic risks. In December 2004, the SEC adopted Rule IA-2333 mandating that hedge fund advisers with more than \$25 million in AUM register with the SEC under the Investment Advisers Act of 1940 (SEC [2004]). Under this rule, hedge fund advisers were subject to SEC inspections and limited disclosure requirements, including Form ADV Part 1 (introduced in 1979 for registered investment advisers).

Rule IA-2333 precipitated strong opposition from hedge fund managers. Hedge fund manager Philip Goldstein sued the SEC, and in 2006 Rule IA-2333 was overturned and vacated as arbitrary by the DC Circuit (SEC [2006]). Hedge fund advisers were then given the option to deregister. The

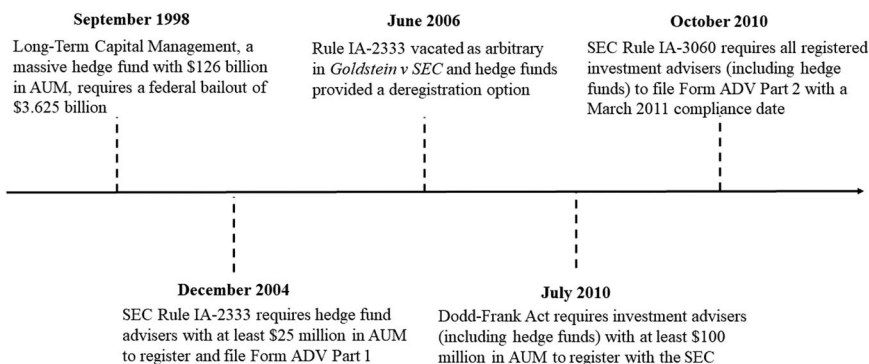


FIG. 2.—Timeline of hedge fund regulations in the United States. This figure outlines the timeline of U.S. hedge fund regulations.

Dodd-Frank Act of 2010 mooted the court's decision by explicitly requiring (subject to a few exemptions) that hedge fund advisers with more than \$100 million in AUM—both U.S. and foreign advisers with U.S. clients—register with the SEC. As a result, registered advisers are required to disclose factors pertaining to their business and operations (SEC [2011c]) and are subject to regulatory examinations and enforcement actions.^{6,7}

There are at least three motivations behind the push for increased hedge fund regulation. First, the SEC is concerned with fraud by investment advisers and asserts that SEC registration reduces hedge funds' opacity and increases regulators' ability to detect fraud and financial misconduct (SEC [2004]). Second, because hedge funds pose systemic risk to the economy, registration requirements under IA-2333 and asset positions and leverage reported on Form PF are used to monitor risks to the U.S. financial system (SEC [2011b]). Finally, hedge fund regulation is believed to protect investors. Although hedge funds traditionally target wealthy accredited individuals and institutional investors outside the normal focus of the SEC, they also manage investments of small investors and pension plans, putting workers' and retirees' capital at risk (Sklar [2008]).⁸

⁶ Regulatory examinations include periodic, special, unscheduled, or other examinations conducted by the SEC's Office of Compliance Inspections and Examinations. See www.sec.gov/about/offices/ocie/ocieoverview.pdf.

⁷ Hedge fund advisers that hold "section 13(f) securities, as defined in rule 13f-1(c), having an aggregate fair market value on the last trading day of any month of any calendar year of at least \$100 million" (SEC [2014]) are also required to disclose their quarterly portfolio holdings. SEC-registered hedge fund advisers with over \$150 million AUM are required to file Form PF (not publicly available) to the Financial Stability Oversight Council (FSOC) for the purpose of risk monitoring.

⁸ SEC commissioner Luis Aguilar stated in a 2013 speech that "the one indispensable fact to remember is that behind all institutional investors and their portfolio managers are millions of American workers, savers, policyholders, retirees, and other individual investors, who rely

2.2 FORM ADV

Registered investment advisers (including hedge fund advisers) are required to file Form ADV and make annual updates within 90 days of the fiscal year-end or make other-than-annual amendments when key information in Form ADV becomes inaccurate. We focus on information contained in Form ADV Part 1A and 2A, as information from the other parts of Form ADV is not relevant for constructing our measure of disclosure inconsistency.⁹ For simplicity, we refer to Part 1A as “Part 1” and Part 2A as “Part 2.” Both parts are generally prepared in-house with varying degrees of involvement by external law firms/compliance consultants. Part 1 has a check-the-box, fill-in-the-blank format and focuses on the hedge fund adviser’s business practices, ownership, clients, financial industry affiliation, client interests, and disciplinary history. Part 1 is reviewed by the SEC and is used to process registrations and manage regulatory and examination programs, although we note that Part 1 is not audited or externally validated and may also contain errors.¹⁰

The SEC passed Rule IA-3060 in October 2010, requiring registered investment advisers to file Form ADV Part 2 with a March 2011 compliance date (see figure 2). Form ADV Part 2 is “designed to promote effective communication” (SEC [2019b]) and is intended to be the primary disclosure document that advisers provide to their clients (SEC [2011a]). Instead of a check-the-box, fill-in-the-blank format, Form ADV Part 2 has a plain-English narrative format that takes into consideration clients’ financial sophistication. Commonly referred to as the “investment adviser brochure,” Form ADV Part 2 includes 18 items where advisers describe their business, fee schedules, risks, COIs, disciplinary history, and other information intended to help clients make informed investment decisions. As outlined in table IA.1, there is extensive overlap in the information provided in Form ADV

on those they entrust with their monies to provide for a safe and secure retirement, to help them save for a home or college education, and to participate in the American dream” (<https://www.sec.gov/news/speech/2013-spch041913laa.htm>). See also <https://fortune.com/2017/07/28/retirement-pensions-hedge-funds/>.

⁹ There are currently three parts to Form ADV. Part 3 (also referred to as Form CRS) pertains only to advisers with retail investors, which typically does not include hedge funds. Form ADV Part 1B is a variant of Part 1A and only applies to advisers registered with a state securities authority. Part 2B is a brochure supplement containing personal information of supervised persons, which not all SEC-registered advisers are required to file.

¹⁰ It is also possible that advisers omit material information from both Parts 1 and 2. As Part 1 is reviewed by the SEC, omissions of material problems in this part are arguably more difficult and costly for the adviser, although discussion with lawyers indicates that it is not always clear what the SEC is asking for and mistakes can happen. See section 5.2 for a discussion of relevant enforcement issues. Hedge funds may also file to expunge past disciplinary histories and legally omit such information, a lengthy and costly process that does not happen frequently (Charoenwong, Kwan, and Umar [2019]).

Parts 1 and 2. Both require disclosures of business practices, clients, types of advisory services offered, disciplinary histories, and COIs.¹¹

The disclosure requirements for the overlapping items of Parts 1 and 2 differ in three aspects. First, Part 2 has a slightly narrower scope. Part 1 requires advisers to disclose disciplinary history and COIs related to the hedge fund adviser and its advisory affiliates,¹² while Part 2 mandates disciplinary history disclosures related to the advisers and its management persons. Second, Part 2 generally requires disclosures of material information. For example, Part 2 requires advisers to “list specific legal and disciplinary events presumed to be material.” The determination of materiality is left to the discretion of advisers.¹³ Third, disclosures in Part 2 are not verified by the SEC. However, the SEC does require that information in the brochure be true and free of omissions of material facts and advises all potential investors to “always ask for, and read carefully, both parts of the adviser’s Form ADV” (SEC [2011a]). One plausible reason for this lack of verification is resource constraints (Kedia and Rajgopal [2011]).

2.3 PUBLIC AND PRIVATE COMMUNICATION

Form ADV is the primary public filing used to communicate with hedge fund investors.¹⁴ Private communication, such as private placement memorandum and investor letters (Cassar et al. [2018]), is unlikely to entirely eliminate the usefulness of public information. Private communication only pertains to established fund-client relationships. Public disclosures, in contrast, allow potential investors to compare across fund products before making their investments, reducing search costs for investment

¹¹ Each disclosure also contains unique information. For instance, unique to Part 1, advisers must disclose specific information about employees (Items 5.A and 5.B) and, for private advisers, disclose minimum investment requirements, the number of beneficiary owners, and auditors (section 7.B.1). Unique to Part 2, advisers must summarize material changes since the last update (item 2) and disclose fee schedules for different types of services or products offered to clients (item 5).

¹² According to the instructions of Part 1, advisory affiliates are: (1) current employees (other than employees performing only clerical, administrative, support or similar functions); (2) officers, partners, or directors (or any person performing similar functions); and (3) all persons directly or indirectly controlling the adviser or controlled by the adviser.

¹³ According to the instructions of Part 2, advisers should consider the following factors when determining whether a disciplinary event is material: “(1) the proximity of the person involved in the disciplinary event to the advisory function; (2) the nature of the infraction that led to the disciplinary event; (3) the severity of the disciplinary sanction; and (4) the time elapsed since the date of the disciplinary event.”

¹⁴ The SEC allows advisers to satisfy disclosure requirements “in [their] brochure or by some other means.” Although we cannot rule out the existence of alternative communication channels, our evidence suggests that these channels are unlikely to be advisers’ primary means of communication. Specifically, if investors are sophisticated enough to understand the implications of these private communications, we would expect to see them respond to inconsistencies. However, we fail to find evidence that investors respond to the implications of inconsistencies (see section 5.1.1).

products. The presence of public disclosures can also enhance the credibility of private communications. The SEC mandates public disclosures to reduce fraud, manage systemic risk (which individual hedge funds do not internalize), and protect investors.

2.4 DATA

We rely on the intersection of three data sources: Form ADV Part 1, Form ADV Part 2, and Thomson Reuter's Lipper TASS Hedge Fund Database. We obtain Form ADV Part 1 data for all SEC-registered investment advisers directly from the SEC's Web site.¹⁵ The SEC provides downloadable spreadsheets (updated monthly) that contain much of Part 1's content going back to 2006. We supplement these data with additional variables (e.g., ownership schedules) collected directly from each adviser's most recent Form ADV Part 1 filing.

We obtain advisers' annual Form ADV Part 2 brochures for the years 2014–2018 from the SEC's Investment Adviser Public Disclosure (IAPD) website and through a Freedom of Information Act (FOIA) request filed with the SEC.¹⁶ We identify brochures using each adviser's unique SEC registration identifier, obtained from Form ADV Part 1.

We collect detailed fund-level information from Thomson Reuter's Lipper TASS Hedge Fund Database for the years 2014 to 2018 and hedge fund manager characteristics from LinkedIn. Detailed variable definitions are provided in appendix A. We merge TASS data with Form ADV data using a name-by-name match and verify matches using adviser locations. This procedure results in a final sample of 570 hedge funds managed by 131 U.S. and 59 foreign advisers. The sample size is comparable to prior studies that examine hedge fund disclosures (e.g., Cassar and Gerakos [2010, 2011], Honigsberg [2019]).

3. *Inconsistent Disclosures*

Our objective is to study whether hedge funds strategically withhold from investors information in Part 2 that has negative performance implications. Withholding negative information potentially helps hedge funds maintain or improve fund flows. Evidence of a correlation between weaker fund performance and such inconsistencies would support this hypothesis. However, as discussed in section 2.2, advisers are only required to disclose material information in Part 2. If most advisers act in good faith (e.g., driven by

¹⁵ <https://www.sec.gov/data-research/sec-markets-data/information-about-registered-investment-advisers-exempt-reporting-advisers>

¹⁶ See <https://adviserinfo.sec.gov/> for advisers' most recent brochures and <https://www.sec.gov/foia-services/frequently-requested-documents/form-adv-data#part2> for historical brochures. We also look for additional adviser-level information (e.g., customer complaints) on brokercheck.finra.org and find that all adviser pages link to the adviser's IAPD page.

reputational concerns) and omit only immaterial information, then inconsistent disclosures would not be indicative of weaker fund performance.

To test our hypothesis, we identify inconsistencies that are more likely due to adviser discretion based on the implication of these disclosures for fund performance and evidence from prior research. Specifically, prior research shows that the presence of COIs and disciplinary histories increases operational risk and has adverse effects on fund performance (Brown et al. [2008, 2012b]). Excessive investment risk also increases the likelihood of deteriorating fund performance. Thus, we examine whether hedge fund advisers strategically omit negative information regarding COIs, disciplinary histories, and limit their discussion of material risks. We discuss each type of disclosure below.

3.1 DISCIPLINARY HISTORIES

Form ADV Part 1 requires that advisers disclose all disciplinary events for the past 10 years involving themselves or an advisory affiliate. Disciplinary events include past regulatory and legal violations, financial-related misdemeanors or charges, and felonies; are useful predictors of fraud (Dimmock and Gerken [2012]); and are associated with operational risk (Brown et al. [2012b]). The presence of these events is referred to as a disciplinary history.

Form ADV Part 1 Item 11 contains 8 sections and 25 questions related to past disciplinary events. Advisers with any disciplinary history are also required to file Disclosure Reporting Pages (DRPs) with their Part 1 filing, which contains additional details about the disciplinary events. For all advisers that answer “yes” to any of these questions, we manually check the adviser’s corresponding Form ADV Part 2 Item 9. Advisers are required to describe in Item 9 all disciplinary events for the past 10 years “that are material to a client’s or prospective client’s evaluation of [the firm’s] advisory business or the integrity of ... management” (SEC [2019b]). Instructions to Form ADV Part 2 state that disciplinary events are presumed to be material, and only if the adviser “rebut[s] the presumption of materiality” and prepares and maintains a memorandum outlining the factors corroborating this rebuttal, does the disciplinary event not have to be disclosed in the brochure (SEC [2019b]). However, these memoranda are not publicly available. Advisers can thus use discretion to omit mention of material past disciplinary histories in their brochures in hopes of improving or maintaining fund flows. We classify a brochure as inconsistent if the hedge fund adviser indicates it has a disciplinary history in Part 1 but explicitly discloses that it has no material disciplinary history in its contemporaneous brochure (*Inconsistent:History*).¹⁷ Table IA.3 provides an anecdote where an adviser answered “yes” to seven disciplinary history questions in its 2018

¹⁷ Table IA.2, panel A, classifies each question in Part 1 Item 11 and tabulates the percentage of all advisers as well as advisers with an inconsistency who answer “yes” to these questions. Inconsistent advisers most commonly indicate in Part 1 that they have a regulatory history in-

Form ADV Part 1, yet indicated in its brochure that it has not been subject to any material legal or disciplinary action in the past 10 years.

The top section of table 1 summarizes advisers' disciplinary history disclosures. During each sample year, 16%–21% of advisers indicate in their Form ADV Part 1 that they have a disciplinary history, yet 41%–43% of these advisers disclose in their corresponding brochure that they have no disciplinary history. Overall, each year, 7%–9% of advisers provide an inconsistent disciplinary history disclosure, and 81%–100% of advisers that have these inconsistencies were also inconsistent the previous year. In comparison, Brown et al. [2012b] find that 16% of hedge funds misreport past disciplinary histories to due diligence firms hired to investigate these funds.

3.2 CONFLICTS OF INTEREST

COIs include a wide range of issues and can be broadly classified into two categories: participation and interest in client transactions, and investment and brokerage discretion. Form ADV Part 1 Item 8 contains 9 sections and 18 questions regarding COIs that might materially affect the advisory relationship. We focus on six uncommon external COIs that correlate with operational risk and predict fraud (Brown et al. [2008], Dimmock and Gerken [2012]). Common COIs often involve issues that are well-known by investors, whereas uncommon COIs are more likely to be material and plausibly deserve additional scrutiny (Honigsberg [2019]). These six uncommon COIs have affirmative responses of 12.3% or less and cover principal transactions, agency cross transactions, brokerage discretion, soft dollar benefits, and client referrals.

Advisers are instructed to “make full disclosure of all material conflicts of interest ... that could affect the advisory relationship” in their brochure (SEC [2019b]) and have a fiduciary duty to make “full and fair disclosure of all conflicts of interest” (SEC [2019a]). For example, Form ADV Part 2 Item 11.B requires advisers to disclose principal transactions with clients (corresponding to Q8A1 in Form ADV Part 1), and Item 12.A1 requires them to disclose “soft dollar benefits” received from related parties (corresponding to Q8G in Form ADV Part 1). Similar to disciplinary histories, advisers can use discretion to omit discussion of material COIs in hopes of maintaining or improving fund flows. We classify a brochure as inconsistent if the adviser indicates one or more of these six uncommon COIs on its Form ADV Part 1 but fails to mention or discuss the COI in its brochure (*Inconsistent:COI*).¹⁸

volving a violation of federal, state, or foreign agency investment-related regulation or statute (59.7%), followed by having one of these agencies place an order against the adviser in connection with an investment-related activity (27.4%).

¹⁸Specifically, we search for descriptions of “principal transactions” or “agency cross transactions” in Item 11 of Form ADV Part 2 if an adviser provides an affirmative response to Q8A1 or Q8B1 in Form ADV Part 1; for descriptions of “brokerage discretion” in Items 10 and 12 of Form ADV Part 2 if an adviser provides an affirmative response to Q8D or Q8F in Form ADV

TABLE 1
Inconsistent Disclosures

	# (%) Hedge Fund Advisers					
	2014 (2)	2015 (3)	2016 (4)	2017 (5)	2018 (6)	
(1)						
<i>Disciplinary History</i>						
Adviser indicates the existence of a past disciplinary event in Part 1 (<i>Disciplinary History</i>)	30 (16%)	36 (19%)	38 (20%)	39 (21%)	40 (21%)	
Subset that disclose no material past disciplinary events in Part 2 (<i>Inconsistent:History</i>)	13 (43%)	15 (42%)	16 (42%)	16 (41%)	17 (43%)	
Subset that were inconsistent in the previous year		13 (87%)	13 (81%)	16 (100%)	16 (94%)	
<i>Uncommon Conflict of Interest (COI)</i>						
Adviser lists an uncommon COI in Part 1 (<i>Uncommon COI</i>)	45 (24%)	45 (24%)	43 (23%)	37 (19%)	35 (18%)	
Subset that make no disclosure in Part 2 of the uncommon COI (<i>Inconsistent:COI</i>)	35 (78%)	33 (73%)	31 (72%)	25 (68%)	24 (69%)	
Subset that were inconsistent in the previous year		32 (97%)	29 (94%)	23 (92%)	22 (92%)	
<i>Risk</i>						
An adviser's fund has standard deviation of returns over previous three years in top tercile (<i>High Risk</i>)	88 (46%)	85 (45%)	91 (48%)	81 (43%)	93 (49%)	
Subset with Part 2 risk discussion section or uncertainty words in bottom tercile (<i>Inconsistent:Risk</i>)	40 (45%)	38 (45%)	44 (48%)	39 (48%)	46 (49%)	
Subset that were inconsistent in the previous year		35 (92%)	32 (73%)	32 (82%)	36 (78%)	
<i>Any Inconsistency</i>						
Adviser has a disciplinary history or uncommon COI per Part 1 or a high risk fund (<i>Problem adviser</i>)	121 (64%)	117 (62%)	121 (64%)	120 (63%)	123 (65%)	
Subset that provide an inconsistent disclosure in Part 2 (<i>Inconsistent:Any</i>)	74 (61%)	72 (62%)	77 (64%)	72 (60%)	76 (62%)	
Subset that were inconsistent in the previous year		67 (93%)	64 (83%)	65 (90%)	67 (88%)	
One type of inconsistency in Part 2 (DH or COI or risk)	62 (84%)	60 (83%)	64 (83%)	64 (89%)	66 (87%)	
<i>Brochure Amendments</i>						
Adviser files a brochure amendment	59 (31%)	46 (24%)	51 (27%)	52 (27%)	32 (17%)	
Subset that provide an inconsistent disclosure in Part 2 (<i>Inconsistent:Any</i>)	25 (42%)	21 (46%)	25 (49%)	21 (40%)	10 (31%)	
Subset whose inconsistency is corrected in the amendments	0 (0%)	0 (0%)	2 (8%)	0 (0%)	0 (0%)	

This table describes inconsistencies in hedge fund advisers' Form ADV Part 2 (brochure). The sample is 190 advisers that were alive as of the end of 2018 with Form ADV Part 1, Part 2 data, and performance data from TASS. Column 1 defines the set of advisers, and columns 2–6 indicate the number and percentage of advisers meeting the criteria each year.

Table IA.2, panel B, lists the six uncommon COIs. The table also tabulates, for each question in Part 1 Item 8, the percentage of all advisers as well as advisers with an inconsistency who answer “yes” to these questions. Among the six uncommon COIs, inconsistent advisers most commonly indicate in Part 1 that they have conflicts related to client referrals (53.5%), proprietary interest (41.6%), and brokerage discretion (40.9%). Table IA.4 provides an anecdote where an adviser acknowledged in Part 1 that agency cross transactions exist yet simultaneously indicated in Part 2 that it does not engage in these transactions.

The second section of table 1 summarizes hedge fund advisers’ uncommon COI disclosures. During each sample year, 18%–24% of advisers indicate in their Form ADV Part 1 that they have at least one of these six COIs, yet 68%–78% of these advisers do not mention or discuss in their corresponding brochure at least one of these COIs. Overall, each year, 13%–18% of advisers provide an inconsistent COI disclosure, and 92%–97% of advisers that have these inconsistencies were also inconsistent the previous year.

3.3 RISK DISCLOSURES

Compared with other investment vehicles such as mutual and index funds, hedge funds typically use leverage and investment strategies that involve material and possibly unusual risks. The SEC requires that advisers use their brochure to (1) inform clients of potential losses that clients should be prepared to bear and (2) explain risks involved in the adviser’s investment strategy as well as significant or unusual risks associated with the type of securities recommended to clients and the frequency of trading (SEC [2019b]). Prior research finds that risk disclosures affect product quality and risk-taking in financial and other markets (Longo [2005], Madsen and McMullin [2020]). Advisers are given latitude in the detail and number of risks discussed in their brochures.

There are no quantifiable measures of risk in Form ADV Part 1. We thus compare discussions of risk in brochures to the historical return volatility of the advisers’ funds (calculated as the standard deviation of reported monthly fund returns over the previous three years), which is observable when the brochure is filed. As the SEC stipulates that advisers adequately *explain* material risks involved (as opposed to simply disclosing the existence of risk), funds with more historically volatile returns should provide a lengthier discussion of their risks to justify the risk. We use two text-based constructs to measure discussion of risk: the length of the mandated risk discussion in Item 8 and the total count of uncertainty-related words (*Inconsistent:Risk*), identified using the Loughran-McDonald sentiment word lists. We scale both measures by the total number of words

Part 1; for descriptions of “soft dollar benefits” in Item 12 of Form ADV Part 2 if an adviser answers “yes” to Q8G1 and “no” to Q8G2 in Form ADV Part 1; and for descriptions of “client referrals” in Item 10 of Form ADV Part 2 if an adviser provides an affirmative answer to Q8I in Form ADV Part 1.

in the brochure to capture the relative importance of risk discussion in the brochure. We classify a brochure as inconsistent if at least one of the advisers' funds has historical return volatility in the top tercile but a discussion of risk in the bottom tercile (using either measure).¹⁹ Table IA.5 reproduces risk sections (one short, one long) from two advisers that have high-risk funds and the same investment style.

The third section of table 1 summarizes hedge fund advisers' risk disclosures. During each sample year, 43%–49% of advisers have a fund with high historical volatility.²⁰ Of these, 45%–49% have a length of the mandated risk discussion or count of uncertainty words in the bottom tercile. Overall, each year, 20%–24% of advisers provide an inconsistent risk disclosure, and 73%–92% of advisers that have these inconsistencies were also inconsistent the previous year.

3.4 ANY INCONSISTENCY

Each of these three inconsistencies plausibly arises through the use of strategic discretion by investment advisers. Each also captures a different aspect of firm performance (operational and investment risk). We thus combine these three measures to flag hedge fund advisers with any type of strategic inconsistency (*Inconsistent:Any*).

Table IA.6 summarizes different groupings of adviser problems and inconsistencies. We find that given the *opportunity* to be inconsistent (i.e., a fund has at least one of the problems we identify), COIs result in the greatest number of inconsistencies, followed by high risk and disciplinary histories. Furthermore, each year, 31%–41% of advisers have multiple inconsistencies, suggesting that the different types of disclosure inconsistencies are correlated constructs.

The fourth section of table 1 indicates that each sample year, 62%–65% of advisers have either a disciplinary history, uncommon COI, or a fund with high historical volatility, and we label them as *problem advisers*. Of these, 60%–64% have at least one inconsistent disclosure. Overall, 38%–41% of advisers provide an inconsistent disclosure, and 88%–93% of advisers that have any inconsistency were also inconsistent the previous year.

Advisers are also required to promptly file an “other-than-annual amendment” when the disclosed information is deemed inaccurate. In the final section of table 1, we find that filing an amendment is fairly common, with

¹⁹ A low discussion of risk could reflect managers' expectations that future performance will be less risky. While possible, 94% of funds in the top tercile of historical return volatility for the 2016–2018 period are also in the top tercile for the 2017–2019 period, suggesting either that these managers hold incorrect beliefs about their expected performance or that the low discussion represents strategic disclosure.

²⁰ Because advisers can have more than one fund, more than 33% of advisers can thus be classified as high risk. Results are qualitatively similar in terms of sign and statistical significance if we calculate historical return volatility at the adviser (rather than fund) level.

17%–31% of advisers filing an amendment each year, and 31%–49% of these advisers have an underlying inconsistency. However, these amendments rarely fix an existing inconsistency, with only two advisers filing amendments that rectify existing inconsistencies.

4. Inconsistencies and Fund Performance

4.1 SUMMARY STATISTICS

Table 2, panel A, reports summary statistics for various fund characteristics using available data for the 570 funds and 190 advisers in our sample. The majority of these variables are constant over our sample period, with the exception of size and age, for which we report the sample average. The average fund-level AUM is \$691.24 million. For brevity, we do not describe all these variables here; see appendix A for the definition and construction of all variables.

Table 2, panel B reports monthly and panel C quarterly summary statistics for the 392 hedge funds (15,825 fund months and 4,184 fund quarters) in our sample with available controls for empirical analyses.²¹ We calculate hedge fund abnormal performance using standard measures in the hedge fund literature (i.e., *Excess return*, *Alpha (CAPM)*, *Alpha (Carhart)*, and *Alpha (Fung and Hsieh)*). We exclude monthly returns greater than 100% or less than –100%, which are likely attributable to reporting errors, and multiply all performance measures by 100 to be in percentage points. The mean (median) seven-factor monthly *Alpha (Fung and Hsieh)* is 0.013 (0.144)%. Quarterly net flows are computed as the difference between a fund's AUM of quarter t and the AUM of quarter $t - 1$ adjusted for the returns earned in quarter t , divided by the AUM of quarter $t - 1$. We exclude observations that have a capital inflow or outflow over 50% of the previous quarter's AUM to eliminate the influence of extreme values and reporting errors in the database. The median quarterly net flow is close to zero.

Panel D reports the sample decomposition by fund style. Funds of hedge funds (28.6%), long/short equity hedge (25.4%), and multistrategy (12.3%) are the three most common strategies used by hedge funds. Options strategy (0.4%), convertible arbitrage (0.7%), and equity market neutral (1.2%) are the three strategies least used by funds in our sample.

4.2 INCONSISTENCIES AND FUND PERFORMANCE

We first examine the relation between inconsistencies and contemporaneous fund performance by estimating the following regression:²²

$$\text{Alpha}_{i,t,m} = \beta \cdot \text{InconsistentAny}_{i,t} + \gamma_1 \cdot \text{Problem adviser}_{i,t} + \gamma_2 \cdot \text{Log size}_{i,t,m}$$

²¹ Of the 570 funds in our initial sample, we lose 157 funds due to missing data on fund size, 6 due to missing data on fund flows, 12 due to missing data on incurred risk, and 1 singleton due to its investment strategy.

²² Our sample only includes funds where the adviser has voluntarily reported fund performance. To evaluate any likely bias from selective reporting, we compare the adviser AUM

TABLE 2
Descriptive Statistics

Panel A: Fund characteristics						
	N	Mean	Standard Deviation	Distribution		
				10th Percentile	Median	90th Percentile
<i>Fund Characteristics</i>						
Size (\$ millions)	413	691.24	3,129.30	5.93	80.12	1,166.86
Age	570	11.59	5.66	5.08	10.54	20.08
Number of funds	570	10.51	14.43	1.00	4.00	29.00
Large family	570	0.24	0.43	0.00	0.00	1.00
Management fees	560	1.39	0.50	0.85	1.50	2.00
Incentive fees	513	15.24	7.28	0.00	20.00	20.00
Highwater mark	570	0.65	0.48	0.00	1.00	1.00
Lockup period	570	94.42	197.01	0.00	0.00	360.00
Redemption notice period	569	49.14	39.76	2.00	37.00	95.00
Total restrictions	569	143.72	211.01	2.00	45.00	425.00
Redemption frequency	536	75.63	83.87	30.00	30.00	90.00
Subscription period	537	32.29	26.08	30.00	30.00	30.00
Minimum investment (\$ millions)	559	4.29	53.10	0.025	0.25	2.00
Payout period	567	16.18	23.40	0.00	10.00	30.00
Open to public	570	0.12	0.32	0.00	0.00	1.00
Foreign	570	0.50	0.50	0.00	0.00	1.00
Fiduciary	570	0.02	0.14	0.00	0.00	0.00
<i>Lenders and Auditors</i>						
Leveraged	570	0.52	0.50	0.00	1.00	1.00
Max leverage	345	73.14	135.08	0.00	0.00	250.00
Avg leverage	345	34.90	72.89	0.00	0.00	140.00
Margin	368	0.31	0.46	0.00	0.00	1.00
Big 4	507	0.72	0.45	0.00	1.00	1.00
<i>Ownership and Clients</i>						
Direct owners	570	2.03	1.86	1.00	1.00	5.00
Indirect owners	570	3.31	3.67	0.00	2.00	8.00
Controlling	570	0.92	0.47	0.00	1.00	1.00
75%Num	570	0.75	0.51	0.00	1.00	1.00
75%Pct	570	0.68	0.45	0.00	1.00	1.00
Domestic direct	570	0.48	0.91	0.00	0.00	1.00
InvestCo	570	0.25	0.43	0.00	0.00	1.00
Pension	570	0.23	0.42	0.00	0.00	1.00
<i>Management</i>						
Fund exp (fund manager)	120	17.43	7.38	8.00	17.00	26.00
Ind exp (fund manager)	100	28.37	8.78	17.00	28.00	41.00
CEO (fund manager)	122	0.66	0.48	0.00	1.00	1.00
Masters degree (fund manager)	114	0.69	0.46	0.00	1.00	1.00
Male (fund manager)	122	0.96	0.20	1.00	1.00	1.00
Fund exp (CCO)	152	10.34	5.52	2.00	10.00	18.00
Ind exp (CCO)	123	22.15	8.16	13.00	22.00	33.00
CEO (CCO)	152	0.06	0.24	0.00	0.00	0.00
Masters degree (CCO)	138	0.59	0.49	0.00	1.00	1.00
Male (CCO)	152	0.66	0.47	0.00	1.00	1.00

(Continued)

TABLE 2—(Continued)

Panel A: Fund characteristics

	N	Mean	Standard Deviation	Distribution		
				10th Percentile	Median	90th Percentile
<i>Misreporting</i>						
Kink	562	0.05	0.21	0.00	0.00	0.00
December spike	562	0.07	0.26	0.00	0.00	0.00
Low index beta	569	0.15	0.36	0.00	0.00	1.00
Low max R^2	569	0.26	0.44	0.00	0.00	1.00
<i>Fund Brochure (at Adviser Level)</i>						
Neg	169	1.78	0.53	1.10	1.75	2.49
Pos	169	0.74	0.25	0.52	0.71	0.95
Uncertain	169	1.32	0.40	0.85	1.29	1.80
Constrain	169	0.74	0.19	0.51	0.72	0.99
Litigious	169	0.68	0.23	0.42	0.66	0.98
Weak	169	0.25	0.10	0.13	0.25	0.38
Mod	169	0.58	0.15	0.39	0.56	0.77
Strong	169	0.70	0.20	0.48	0.69	0.97

Panel B: Fund-month summary statistics

	<i>N</i>	Mean	Standard Deviation	Distribution		
				10th Percentile	Median	90th Percentile
<i>Performance Measures</i>						
Excess return	15,825	0.21	3.16	−2.84	0.32	3.14
Alpha (CAPM)	15,825	0.04	2.95	−2.76	0.14	2.69
Alpha (Carhart)	15,825	−0.03	3.17	−3.04	0.06	2.80
Alpha (Fung and Hsieh)	15,825	0.01	3.93	−3.52	0.14	3.38
<i>Measures of Inconsistency</i>						
Inconsistent:Any	15,825	0.41	0.49	0.00	0.00	1.00
Inconsistent:Risk	15,825	0.20	0.40	0.00	0.00	1.00
Inconsistent:COI	15,825	0.20	0.40	0.00	0.00	1.00
Inconsistent:History	15,825	0.09	0.29	0.00	0.00	0.00
Inconsistent:Business	15,825	0.88	0.33	0.00	1.00	1.00
Inconsistent:Other	15,825	0.30	0.46	0.00	0.00	1.00
Inconsistent:Affiliations	15,825	0.45	0.50	0.00	0.00	1.00
Inconsistent:Custody	15,825	0.25	0.43	0.00	0.00	1.00
<i>Controls</i>						
Problem adviser	15,825	0.71	0.45	0.00	1.00	1.00
Large family	15,825	0.14	0.34	0.00	0.00	1.00
Size (\$ millions)	15,825	879	4,120	5	92	1,406
Log size	15,825	18.25	2.23	15.50	18.34	21.06

Panel C: Fund-quarter summary statistics

	N	Mean	Standard Deviation	Distribution		
				10th Percentile	Median	90th Percentile
Fund flow	4,184	−0.02	0.09	−0.11	−0.01	0.05
Frank1	4,184	0.27	0.10	0.08	0.33	0.33

(Continued)

TABLE 2—(Continued)

Panel C: Fund-quarter summary statistics						
	N	Mean	Standard Deviation	Distribution		
				10th Percentile	Median	90th Percentile
Frank2	4,184	0.17	0.15	0.00	0.18	0.33
Frank3	4,184	0.07	0.11	0.00	0.00	0.26
Inconsistent:Any	4,184	0.42	0.49	0.00	0.00	1.00
Problem adviser	4,184	0.72	0.45	0.00	1.00	1.00

Panel D: Fund styles				
Primary Category	Full Sample	%	Regression Sample	%
Fund of funds	163	28.6	116	29.6
Long/short equity hedge	145	25.4	97	24.7
Multistrategy	70	12.3	35	8.9
Other	69	12.1	52	13.3
Event driven	52	9.1	38	9.7
Emerging markets	27	4.7	20	5.1
Global macro	22	3.9	18	4.6
Fixed-income arbitrage	9	1.6	6	1.5
Equity market neutral	7	1.2	6	1.5
Convertible arbitrage	4	0.7	4	1.0
Options strategy	2	0.4	0	0.0
Total	570	100	392	100

This table provides sample summary statistics. Panel A presents summary statistics for fund characteristics. Panels B and C present summary statistics for the sample of fund-month and fund-quarter observations covering the period 2014–2018. Panel D summarizes fund styles. See appendix A for the definition and calculation of these variables.

+ γ₃ · Large family_{*i*} + ϕ_{Style×Month} + ε_{*i,t,m*}, (1)

where *Alpha*_{*i,t,m*} is a measure of fund *i*’s performance in month *m* of year *t* and *Inconsistent:Any*_{*i,t*} is an indicator variable for the presence of any strategic inconsistency in the fund adviser’s year *t* brochure. *Problem adviser*_{*i,t*} is an indicator variable that equals one if a fund has an adviser in year *t* with a disciplinary history or uncommon COI per its Form ADV Part 1 or a fund with a standard deviation of returns over the previous three years in the top tercile.

Our primary coefficient of interest (β) is expected to be negative if inconsistent funds perform worse than otherwise comparable problem funds that discuss these issues in their brochures (i.e., consistent funds). The coefficient on *Problem adviser* captures the performance of consistent funds,

from TASS (aggregated across all reported funds) with the adviser AUM per Form ADV, and find that advisers with inconsistencies have on average a lower percentage of AUM in TASS. This selection biases against our findings of lower performance for inconsistent funds if the performance of poor-performing funds is more likely not disclosed.

relative to otherwise comparable non-problem funds (i.e., funds of advisers with no disciplinary history or uncommon COI per their Form ADV Part 1 and no fund with a high historical volatility). We also include as controls *Log size* and *Large family*. As fund investment styles are associated with asset illiquidity (Cassar and Gerakos [2011]), which can affect fund performance (Boyson, Stahel, and Stulz [2010]), we include fund-style-times-month fixed effects, which account for comovement of performance within the same fund style and month (Boyson, Stahel, and Stulz [2010]). We cluster standard errors by investment adviser (a more conservative approach than clustering by fund) because our main explanatory variable of interest varies by investment adviser and fund performance is plausibly correlated among funds of the same adviser.

Table 3 presents estimates of equation (1) using four measures of fund performance: *Excess return* (columns 1–2), *Alpha (CAPM)* (columns 3–4), *Alpha (Carhart)* (columns 5–6), and *Alpha (Fung and Hsieh)* (columns 7–8). All performance measures are multiplied by 100. We first omit *Inconsistent: Any* to examine differences in the performance of all funds with a problem adviser. The coefficient on *Problem adviser* is negative in all odd-numbered columns and statistically significant in columns 3 and 5, suggesting that funds with these problems are associated with poorer performance, plausibly due to higher operational risk (Brown et al. [2008]). The performance for these funds is lower by 12 to 20 basis points, comparable to results in Brown et al. [2008] that funds with disciplinary histories have performance that is lower by 14 basis points.

We next include *Inconsistent: Any* and find that the poorer performance of problem funds concentrates in the subset with inconsistent disclosures. Across all four performance measures, we find that inconsistent funds have worse performance than both consistent funds as well as non-problem funds. Moreover, the coefficient on *Problem adviser* is positive and insignificant in all four specifications, suggesting that consistent funds have performance similar to non-problem funds. Thus it is not the presence of a fund characteristic (e.g., disciplinary history or COI) per se, but rather the decision not to disclose this information, that explains poor performance. In terms of economic magnitudes, inconsistent funds realize monthly abnormal returns that are lower by 26 to 56 basis points, or an annualized 3.1–6.7 percentage points.

Examining the control variables, we find that performance is increasing in fund size. We also estimate a modified version of equation (1) using *Alpha (Fung and Hsieh)* that allows β to vary by year. Figure 3 shows that inconsistent funds have significantly lower performance in three of our four sample years.²³

These results are robust to various research design and sample selection choices. We find similar results if we (1) include fund fixed effects and con-

²³ We exclude the year 2014 from this analysis as we lose the first nine months to estimate *Alpha (Fung and Hsieh)*.

TABLE 3
Fund Performance

	Excess Return		Alpha (CAPM)		Alpha (Carhart)		Alpha (Fung and Hsieh)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Inconsistent:Any		-0.26*** (0.10)		-0.37*** (0.12)		-0.35*** (0.12)		-0.56*** (0.14)
Problem adviser	-0.12 (0.08)	0.04 (0.11)	-0.19* (0.10)	0.03 (0.13)	-0.20** (0.10)	0.01 (0.13)	-0.15 (0.12)	0.19 (0.15)
Log size	0.03* (0.02)	0.03* (0.02)	0.05** (0.02)	0.04** (0.02)	0.05** (0.02)	0.04** (0.02)	0.04* (0.02)	0.04* (0.02)
Large family	0.07 (0.06)	0.04 (0.10)	0.16 (0.13)	0.11 (0.15)	0.20* (0.11)	0.16 (0.14)	0.06 (0.14)	-0.01 (0.18)
Observations	15,825	15,825	15,825	15,825	15,825	15,825	15,825	15,825
Adj <i>R</i> -squared	0.18	0.18	0.10	0.10	0.09	0.09	0.12	0.12
Style × month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

This table analyzes the relation between hedge fund monthly performance and disclosure inconsistency. The dependent variable is as follows: in columns 1–2, the fund’s excess return; in columns 3–4, the CAPM alpha; in columns 5–6, the Carhart alpha; and in columns 7–8, the Fung and Hsieh (2001, 2004) seven-factor alpha. All performance measures are multiplied by 100. The explanatory variable of interest is *Inconsistent:Any_{*i,t*}*, an indicator if fund *i* has an adviser with an inconsistency in their year *t* Form ADV Part 2. We include as additional controls *Problem adviser*, an indicator if fund *i* has an adviser with disciplinary history or uncommon COI per their year *t* form ADV Part 1 or if the adviser has a fund with a standard deviation of monthly returns over the previous three years in the top tercile; *Log size*, the natural log of fund AUM; and *Large family*, an indicator if the fund belongs to a fund family with more than 10 funds. We also include fund-style × month fixed effects. The sample is 392 funds with return and control variable data for the sample period 2014–2018. Standard errors (clustered by adviser) are in parentheses, and *, **, and *** indicate significance at the 0.1, 0.05, and 0.01 level, respectively.

$$\text{Alpha}_{i,t,m} = \alpha + \beta_1 \text{Id}_i + \gamma \cdot \text{Controls}_{i,t,m} + \phi_{\text{Style} \times \text{Month}} + \epsilon_{i,t,m}.$$

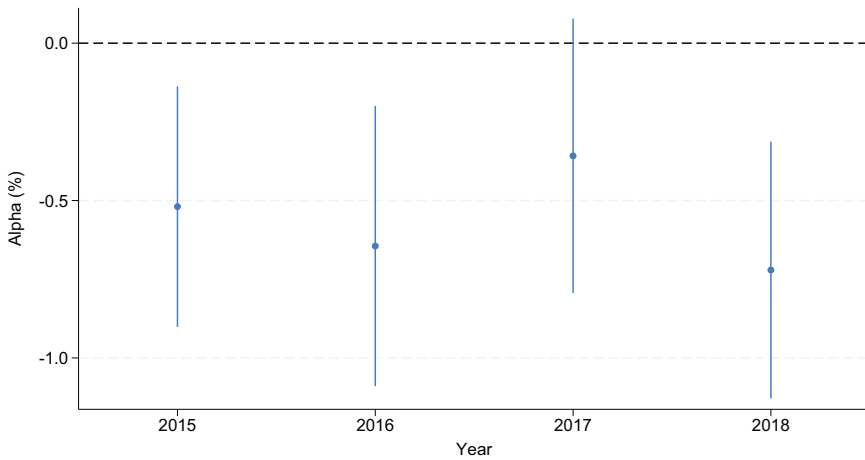


FIG. 3.—Fund returns by year. This figure plots the relation between *Inconsistent:Any* and fund performance by year for the sample period 2015–2018. The dependent variable is *Alpha* (Fung and Hsieh). We estimate the regression below and plot the β coefficient estimates (circles) and corresponding 95% confidence intervals (bars). We include as additional controls *Problem adviser*, *Log size*, and *Large family* as well as fund-style $\times$ month fixed effects. Standard errors are clustered by adviser. $\text{Alpha}_{i,t,m} = \alpha + \sum_{j \in \{2015, \dots, 2018\}} \beta_j \cdot \text{Inconsistent:Any}_{i,t} \times \text{Year}_j + \gamma \cdot \text{Controls}_{i,t,m} + \phi_{\text{Style} \times \text{month}} + \epsilon_{i,t,m}$.

trol for trading restrictions, (2) cluster standard errors by adviser and year-month (or alternatively only by year-month, untabulated), (3) cluster standard errors by fund, (4) use Fama-MacBeth regressions to address cross-sectional correlation in the residuals (e.g., Conley, Gonçalves, and Hansen [2018], Johnson, Kim, and So [2020]), (5) exclude advisers affiliated with mutual funds, (6) exclude funds with a “fund of funds” strategy, or (7) replace *Inconsistent:Any* and *Problem adviser* with the log number of inconsistencies and problems (see table IA.7). This last result also shows that performance is not decreasing in the number of problems (i.e., positive coefficients on the number of problems), but that funds with multiple inconsistencies realize even lower performance than funds with one inconsistency. Results are also robust if we conduct the fund performance analysis at the fund level using the fund’s average performance over the entire sample period (see table IA.8); include funds that deregister from the SEC between 2014 and 2018 to mitigate survival bias concerns (see table IA.9); and use inverse probability weights to address the concern that some advisers have more funds (see table IA.10). We also find that the lower performance of inconsistent funds is present for sample splits by fund size and fund family size (see table IA.11).

The negative association between inconsistencies and performance suggests that advisers strategically omit information with negative performance implications. We rule out several alternative explanations, including that funds are sloppy or overconfident, or that inconsistencies reflect past prob-

lems, and also find that inconsistencies are predictive of future negative performance (see section 6.1). Our findings are consistent with studies of discretionary disclosure in the corporate setting, which find that poor-performing firms are less likely to disclose. An implication of these findings is that the existence of disciplinary histories and COIs for consistent funds is arguably a poor measure of operational risk.

5. *Mechanisms*

In the previous section, we find that advisers who provide inconsistent disclosures tend to underperform, suggesting these omissions reflect economically meaningful issues. This finding raises a natural question: Why are advisers able to omit negative information if their investors can observe the inconsistency? According to disclosure theory, if investors understand that these omissions imply negative future performance, strategic omissions should not help advisers maintain or improve fund flows, discouraging them from engaging in such behavior (Grossman [1981], Milgrom [1981]). In this section, we examine two mechanisms that may explain this disconnect: (1) limited strategic thinking by investors, which can prevent this unraveling effect, and (2) low regulatory enforcement, which reduces the cost of nondisclosure.

5.1 LIMITED STRATEGIC THINKING

Prior research finds that economic agents are often subject to limited strategic thinking and underreact to the implications of strategic actions (known as “cursed equilibrium”) (Eyster and Rabin [2005]). This underreaction occurs because inferring the implications of strategic actions is a complicated process, requiring economic agents to understand the reasons behind these actions and correctly assign probabilities to each reason. Many experimental studies, including strategic disclosure games, find behavior consistent with cursed equilibrium (e.g., Jin, Luca, and Martin [2021]). Although the sophisticated investors in our setting presumably observe the inconsistencies, it can be difficult to evaluate whether they arise from strategic omissions by fund managers or because the omitted information is immaterial. This challenge arises in part because SEC regulation requires only material information to be disclosed in Part 2 and advisers are not required to provide sufficient context for investors to evaluate whether omitted information is material. If disclosures are inconsistent, investors’ default belief might be that the omitted information is immaterial, resulting in an underreaction to the inconsistencies. A failure by investors to appropriately respond to inconsistencies enables advisers’ use of discretion.

5.1.1. Flow-Performance Sensitivity. We first evaluate whether investors’ trading behavior reflects a correct understanding of the implications of disclosure inconsistency. Prior research finds that hedge fund investors tend to chase past performance, which manifests as a positive fund flow-

performance relation, and that fund characteristics can affect that relation (e.g., Brown et al. [2008, 2012b], Goldie, Li, and Masli [2018], Franzoni and Giannetti [2019]). If investors understand the negative relation between inconsistencies and future fund performance, then inconsistent funds should be less likely to attract new capital (flows). In addition, their flows should be potentially more sensitive to past negative performance and less sensitive to past positive performance, as positive (negative) past performance is potentially less (more) likely to be persistent for inconsistent funds. Alternatively, if managers can successfully de-emphasize fund problems by being inconsistent, then fund flow and flow-performance sensitivity will not differ for inconsistent funds.

To test whether disclosure inconsistency mediates either the flow of capital or the flow-performance relation, we follow Dimmock and Gerken [2015] and estimate the following regression:

$$\begin{aligned}
 \text{Flow}_{i,t,q+1} = & \alpha + \sum_{f \in 1,2,3} \delta_f \cdot \text{FRANK}f_{i,t,q} \\
 & + \sum_{f \in 1,2,3} \gamma_f \cdot \text{FRANK}f_{i,t,q} \cdot \text{Problem adviser}_{i,t-1} \\
 & + \sum_{f \in 1,2,3} \beta_f \cdot \text{FRANK}f_{i,t,q} \cdot \text{Inconsistent:Any}_{i,t-1} \\
 & + \eta_1 \cdot \text{Problem adviser}_{i,t-1} + \eta_2 \cdot \text{Inconsistent:Any}_{i,t-1} \\
 & + \zeta \cdot \text{Controls}_{i,t,q} + \phi_{\text{Style} \times \text{Quarter}} + \epsilon_{i,t,q}, \tag{2}
 \end{aligned}$$

where the dependent variable is the quarterly net flow for fund i in quarter $q + 1$ of year t and FRANK is the fund's style-adjusted fractional return rank in quarter q .

We follow prior research (e.g., Brown et al. [2008], Dimmock and Gerken [2015]) and use a piecewise linear specification that allows the flow-performance sensitivity to vary by performance tercile, as prior research finds that the flow-performance relation is stronger for higher ranked funds. The coefficient on FRANK1 captures the flow-performance sensitivity for funds in the bottom performance tercile, and the coefficients on FRANK2 and FRANK3 capture any *incremental* change in that relation for funds in the middle and top terciles, respectively. The sum of the FRANK1 and FRANK2 (FRANK1 , FRANK2 , and FRANK3) coefficients equals the total flow-performance sensitivity for funds in the middle (top) tercile. Interactions between these fractional ranks and $\text{Inconsistent:Any}_{i,t-1}$ and $\text{Problem adviser}_{i,t-1}$ capture differences in the flow-performance sensitivity for these fund types. We also include $\text{Problem adviser}_{i,t-1}$ and $\text{Inconsistent:Any}_{i,t-1}$ to test whether these funds differ in their ability to attract flows. Following prior research (Dimmock and Gerken [2015], Liang et al. [2019]), we include controls for fund flow in quarter q ; standard deviation of returns in quarter q ; indicators for open to public, high watermarks, and leverage;

fees; fund size; and lockup, redemption, and subscription period; as well as fund-style-times-quarter fixed effects. Standard errors are clustered by adviser.

Table 4 presents estimates of equation (2). We find no evidence of a difference in either quarterly flows or the sensitivity of fund flows to performance for inconsistent funds. Column 1 estimates a baseline specification, column 2 includes lag fractional ranks and their interaction terms, columns 3 and 4 estimate Fama-MacBeth quarterly regressions, and column 4 uses fractional rankings across all funds (rather than style adjusted). The coefficient on *FRANK2* is positive and statistically significant in columns 1–4, consistent with increased flow-performance sensitivity for funds in the middle tercile. The evidence is consistent with prior research that investors tend to chase past performance, but there is no evidence that investors are aware or care about the lower future performance of inconsistent funds (i.e., insignificant coefficients on *Inconsistent:Any* and its interaction terms).

Investors' failure to respond to inconsistencies might occur because of trading frictions. Related research finds that short selling restrictions in equity markets can prevent negative news from being reflected in the stock price (Miller [1977], Zhou and Zhou [2020], Fischer, Kim, and Zhou [2022]). Using a determinants model, we find that funds with more trading restrictions are more likely to be inconsistent (see section *Characteristics of Inconsistent Funds* of the online appendix). We thus examine whether investors are more likely to respond to inconsistencies when they face low trading restrictions. We find in table 4, panel B, that investors do not properly respond to the implications of inconsistencies even when they face low trading restrictions (i.e., when redemption notice period is below the median). Additionally, we also link inconsistencies to fund flows either for the subsequent year (rather than only the subsequent quarter) or for quarter $q + 2$, as most trading frictions are only in place for a limited time (e.g., 75% of our sample funds have a total restriction of 95 days or below). Results in table IA.12 suggest that investors do not seem to respond to the implications of the inconsistencies over either of these longer horizons.

5.1.2. Sophisticated Market Participants. We next evaluate whether specific subsets of sophisticated market participants respond to disclosure inconsistency. Because the outcome variables of interest (i.e., the presence of a sophisticated market participant) do not vary across the sample period, we test whether participation differs for inconsistent funds using the following equation:

$$\begin{aligned} \text{Characteristic}_i = & \beta_1 \cdot \text{Inconsistent:Any-max}_i + \beta_2 \cdot \text{Problem adviser-max}_i \\ & + \gamma_1 \cdot \text{Avg log size}_i + \gamma_2 \cdot \text{Large family}_i + \phi_{\text{Style}} + \epsilon_i. \end{aligned} \quad (3)$$

The dependent variable is a fund-level characteristic, *Inconsistent:Any-max* is an indicator equal to one if the fund's adviser was inconsistent in any of the sample years, and *Problem adviser-max* is similarly defined for funds with a problem adviser in any year.

TABLE 4
Flow-Performance Sensitivity

Panel A: Full sample				
	(1)	(2)	(3)	(4)
Inconsistent:Any	-0.01 (0.01)	0.00 (0.01)	0.01 (0.01)	-0.00 (0.01)
Inconsistent:Any × FRANK1	0.02 (0.05)	0.03 (0.05)	-0.05 (0.06)	0.06 (0.07)
Inconsistent:Any × FRANK2	-0.04 (0.04)	-0.04 (0.04)	0.01 (0.07)	-0.06 (0.07)
Inconsistent:Any × FRANK3	0.03 (0.04)	0.04 (0.05)	0.01 (0.05)	-0.00 (0.04)
Problem adviser	-0.02 (0.02)	-0.02 (0.02)	-0.04* (0.02)	-0.03 (0.02)
Problem adviser × FRANK1	0.10 (0.06)	0.10 (0.07)	0.10 (0.07)	0.06 (0.07)
Problem adviser × FRANK2	-0.03 (0.04)	-0.03 (0.04)	-0.07 (0.05)	-0.02 (0.04)
Problem adviser × FRANK3	0.05 (0.04)	0.06 (0.04)	0.08 (0.05)	0.05 (0.05)
FRANK1	-0.10* (0.05)	-0.11** (0.05)	-0.06 (0.06)	-0.07 (0.06)
FRANK2	0.07** (0.03)	0.07** (0.03)	0.07** (0.02)	0.06* (0.03)
FRANK3	-0.03 (0.03)	-0.04 (0.03)	-0.02 (0.04)	0.01 (0.05)
Observations	4,184	4,184	4,184	4,184
Adj <i>R</i> -squared/ Avg <i>R</i> -squared	0.06	0.07	0.25	0.24
Controls	Yes	Yes	Yes	Yes
Style × quarter fixed effects	Yes	Yes	No	No
Style fixed effects	No	No	Yes	Yes
Lag FRANK & interactions	No	Yes	Yes	Yes
Method	Pooled	Pooled	FMB	FMB
Style-adjusted returns	Yes	Yes	Yes	No
Table 4 Panel B: Low trading restrictions				
	(1)	(2)	(3)	(4)
Inconsistent:Any	-0.02* (0.01)	-0.00 (0.02)	0.00 (0.03)	-0.03 (0.03)
Inconsistent:Any × FRANK1	0.07 (0.05)	0.08 (0.05)	0.03 (0.08)	0.22** (0.10)
Inconsistent:Any × FRANK2	-0.02 (0.06)	-0.01 (0.06)	0.06 (0.07)	0.01 (0.08)
Inconsistent:Any × FRANK3	0.01 (0.07)	-0.01 (0.07)	-0.05 (0.09)	-0.07 (0.12)
Problem adviser	-0.02 (0.03)	-0.04 (0.04)	-0.03 (0.03)	-0.01 (0.05)
Problem adviser × FRANK1	0.02 (0.10)	-0.00 (0.10)	0.03 (0.07)	-0.01 (0.15)
Problem adviser × FRANK2	0.05 (0.06)	0.05 (0.06)	-0.05 (0.06)	-0.08 (0.09)
Problem adviser × FRANK3	0.02 (0.07)	0.04 (0.07)	0.12 (0.18)	-0.29 (0.37)

(Continued)

TABLE 4—(Continued)

Table 4 Panel B: Low trading restrictions				
	(1)	(2)	(3)	(4)
FRANK1	−0.09 (0.09)	−0.09 (0.09)	−0.08 (0.08)	−0.12 (0.15)
FRANK2	−0.01 (0.05)	−0.01 (0.05)	0.00 (0.05)	0.04 (0.07)
FRANK3	−0.00 (0.04)	−0.01 (0.04)	−0.02 (0.13)	0.36 (0.34)
Observations	1,881	1,881	1,881	1,881
Adj <i>R</i> -squared/ Avg <i>R</i> -squared	0.04	0.03	0.40	0.41
Controls	Yes	Yes	Yes	Yes
Style × quarter fixed effects	Yes	Yes	No	No
Style fixed effects	No	No	Yes	Yes
Lag FRANK & interactions	No	Yes	Yes	Yes
Method	Pooled	Pooled	FMB	FMB
Style-adjusted returns	Yes	Yes	Yes	No

This table analyzes the relation between quarterly fund flows, past performance, and *Inconsistent:Any*. The dependent variable, $Flow_{i,t,q+1}$, is the net flows into fund i during quarter $q + 1$. The primary explanatory variables are the fractional rank of fund i for quarter q (*Frank1*, *Frank2*, *Frank3*); *Inconsistent:Any_{i,t−1}*, an indicator variable if fund i had an inconsistent disclosure in the previous calendar year; and the interaction of these variables. We also include *Problem adviser_{i,t−1}* and its interaction with the fund's fractional rank. Following Dimmock and Gerken [2015], we include controls for flows in quarter q , standard deviation of fund returns, *Open to public*, *Highwater mark*, *Leveraged*, *Management fees*, *Incentive fees*, *Lockup period*, *Redemption notice period*, *Redemption frequency*, *Subscription period*, and *Size*. See appendix A for variable definitions. Panel A includes all funds with available controls for the 2015–2019 sample period, and panel B includes the subset of funds whose redemption notice period is below the sample median. For each panel, in columns 1 and 2, we include fund-style × quarter fixed effects and estimate pooled quarterly regressions, clustering standard errors by adviser; in columns 3 and 4, we include style fixed effects and estimate Fama-MacBeth regressions. Columns 2–4 include fractional ranks in quarter $q − 1$ and their respective interaction terms. Columns 1–3 use style-adjusted fractional ranks, and column 4 uses raw fractional ranks. Standard errors are in parentheses, and *, **, and *** indicate significance at the 0.1, 0.05, and 0.01 level, respectively.

$$\begin{aligned}
 Flow_{i,t,q+1} = & \alpha + \sum_{f \in 1,2,3} \delta_f \cdot FRANK f_{i,t,q} + \sum_{f \in 1,2,3} \gamma_f \cdot FRANK f_{i,t,q} \cdot Problem\ adviser_{i,t-1} \\
 & + \sum_{f \in 1,2,3} \beta_f \cdot FRANK f_{i,t,q} \cdot Inconsistent:Any_{i,t-1} + \beta_4 \cdot Problem\ adviser_{i,t-1} \\
 & + \beta_5 \cdot Inconsistent:Any_{i,t-1} + \zeta \cdot Controls_{i,t,q} + \phi_{Style \times Quarter} + \epsilon_{i,t,q}.
 \end{aligned}$$

We first examine four lender-related variables: *Leveraged*, an indicator variable that equals one if the fund uses leverage; *Avg leverage* and *Max leverage*, the average and maximum leverage the fund reported to TASS; and *Margin*, an indicator variable that equals one if the fund uses margin to leverage. Greater leverage indicates access to credit via lenders such as prime brokers (Brown et al. [2008]). If lenders are concerned about the potential problems associated with inconsistent funds and are able to anticipate their lower future performance, then we expect β_1 to be negative for all four characteristics.

The results, presented in table 5, panel A, columns 1–4, suggest that lenders are either unaware of or not concerned with the problems associated with inconsistent funds. Examining the first row, we find that

TABLE 5
Sophisticated Market Participants

Panel A: Lenders and auditors	Dependent Variable				
	(1) Leveraged	(2) Max Leverage	(3) Avg Leverage	(4) Margin	(5) Big 4
Inconsistent:Any-max	0.21*** (0.06)	-13.49 (22.28)	-4.78 (12.67)	-0.04 (0.07)	-0.01 (0.06)
Problem adviser-max	-0.22*** (0.08)	36.89 (27.78)	6.96 (14.31)	-0.04 (0.08)	0.33*** (0.08)
Avg log size	0.00 (0.01)	3.49 (3.97)	4.19* (2.29)	-0.03** (0.01)	0.04** (0.01)
Large family	0.43*** (0.09)	10.24 (23.81)	16.85 (15.77)	0.55*** (0.10)	0.49*** (0.08)
Observations	399	274	274	296	349
Adj <i>R</i> squared	0.12	0.11	0.11	0.13	0.23
Style fixed effects	Yes	Yes	Yes	Yes	Yes
Problem adviser + Inconsistent = 0 (<i>p</i> -value)	0.79	0.29	0.87	0.23	0.00

(Continued)

TABLE 5—(Continued)

	Dependent Variable							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Direct Owners	Indirect Owners	Controlling	75%num	75%pct	Domestic Direct	InvestCo	Pension
Inconsistent:Any-max	-0.02 (0.33)	0.28 (0.65)	-0.17 (0.12)	-0.09 (0.13)	0.00 (0.09)	-0.10 (0.17)	0.15* (0.08)	0.06 (0.08)
Problem adviser-max	-0.26 (0.39)	1.00 (0.63)	0.03 (0.16)	0.17 (0.17)	0.10 (0.11)	0.07 (0.21)	-0.14 (0.10)	-0.03 (0.09)
Log size	0.15* (0.09)	0.34* (0.18)	-0.01 (0.02)	0.03 (0.03)	0.00 (0.02)	0.09** (0.04)	-0.03 (0.02)	-0.03 (0.02)
Large family	-2.37*** (0.65)	-0.01 (1.61)	0.21 (0.24)	0.32 (0.24)	0.55*** (0.12)	-0.89** (0.38)	0.25 (0.25)	0.34 (0.24)
Observations	169	169	169	169	169	169	169	169
Adj <i>R</i> -squared	0.03	0.03	-0.06	-0.05	-0.02	0.02	0.04	0.07
Style fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Problem + Inconsistent (<i>p</i> -value)	0.44	0.06	0.22	0.52	0.35	0.88	0.89	0.71

This table analyzes the relation between sophisticated market participants and *Inconsistent:Any-max*, an indicator variable equal to one if fund *i* has an inconsistent disclosure in any of our sample years. The dependent variable is indicated at the top of each column. In panel A, we analyze fund characteristics that reflect the actions of lenders and auditors, and in panel B, we analyze investment adviser ownership characteristics that reflect the actions of equity investors and sophisticated clients. We include as controls *Problem adviser-max*, *Avg log size*, and *Large family* as well as fund-style fixed effects. We tabulate in the table notes the *p*-value for the sum of the coefficients on *Inconsistent:Any-max* and *Problem adviser-max* differs from zero. Robust standard errors are in parentheses, and *, **, and *** indicate significance at the 0.1, 0.05, and 0.01 level, respectively.

$$\text{Characteristic}_i = \alpha + \beta_1 \cdot \text{Inconsistent:Any-max}_i + \gamma \cdot \text{Controls}_i + \phi_{\text{style}} + \epsilon_i.$$

inconsistent funds are more likely to use leverage relative to consistent funds, although they do not differ in the degree of leverage used or the use of margin. An *F*-test indicates that the sum of the coefficients on *Inconsistent:Any-max* and *Problem adviser-max* in column 1 is statistically insignificant, indicating that inconsistent funds' use of leverage also resembles non-problem funds' usage.

We next examine actions by auditors and use as the dependent variable *Big 4*, an indicator equal to one if the fund is audited by a Big 4 accounting firm. As discussed by Brown et al. [2012b], "the auditing firm typically pre-screens managers for the potential risk they pose to the firm before taking them as a client" (p. 225). Brown et al. [2012b] find that the presence of a high-quality auditor is associated with reduced operational risk, suggesting that inconsistent funds may be less able to retain such firms. We find in column 5 that the likelihood that inconsistent funds have a Big 4 accounting firm as their auditor is not statistically different from consistent funds. Furthermore, inconsistent funds are more likely to have a Big 4 auditor compared to non-problem funds. Disclosure inconsistency does not appear to reduce the willingness of a reputable Big 4 firm to audit these funds.

Next, we evaluate six variables that capture actions by the investment company's equity investors: *Direct owners*, the number of direct owners; *Indirect owners*, the number of indirect owners; *Controlling*, the number of controlling owners; *75%num*, the number of direct owners who own at least 75% of the company; *75%pct*, the percentage of direct owners who own at least 75% of the company; and *Domestic direct*, the number of domestic corporations listed as direct owners.²⁴ Prior research suggests that funds sell equity stakes (cash flow rights) to external financial institutions for a variety of reasons: to provide a certification signal to investors that will potentially increase AUM (and fees charged), improve fund visibility, and improve fund governance and monitoring (Mullally [2022]). If outside institutions are concerned with the potential problems associated with inconsistent funds, then we expect there will be fewer direct and indirect owners and that fund ownership will be more concentrated.

In table 5, panel B, columns 1–6, we find that inconsistent funds insignificantly differ from consistent funds in their ownership structure. Our evidence suggests that equity investors are also either unaware of or not concerned with the potential problems associated with inconsistent funds. An *F*-test on the sum of *Inconsistent:Any-max* and *Problem adviser-max* indicates that inconsistent funds have marginally more indirect owners than non-problem funds (*p*-value 0.06).

Finally, we examine the actions of hedge fund clients. Clients such as investment companies and pension funds, which often adopt long-term

²⁴ Prior research suggests that direct ownership by a domestic entity suggests the fund is structured as a venture or partnership and that fear of expropriation makes management more concentrated (Brown et al. [2009], Mullally [2022]), whereas the distinction between direct and indirect owners predominately reflects tax and liability issues.

investment strategies, may be more diligent in selecting and monitoring their hedge fund investments and pay greater attention to potential adviser problems, and thus may be less willing to work with inconsistent funds. In columns 7–8, we investigate the likelihood that the fund has an investment company or pension fund as a client and find that inconsistent funds are marginally *more* likely to have investment company clients and do not differ in their likelihood of having pension fund clients.

Overall, equity investors, clients, lenders, and reputable auditors seem to be unconcerned with the implications of inconsistencies for future fund performance, consistent with limited strategic thinking leading to underreactions to the implications of strategic actions. As discussed above, due to limited strategic thinking, market participants likely believe that the omitted information is immaterial. Fund managers, whose compensation depends on fund flows, can thus successfully de-emphasize fund problems without sacrificing fund flows or deterring market participants.

5.2 LOW LITIGATION RISK

High litigation risk should deter advisers from omitting material information. In a 2017 risk alert, the SEC flagged the use of “inaccurate disclosures” by investment advisers and described it as one of “the five most frequent compliance topics” (SEC [2017]). However, the prevalence of inconsistent disclosures suggests that some advisers view this specific litigation risk as relatively low.

This section documents the limited efforts by the SEC to enforce consistent disclosures as a contributing factor to low litigation risk. We hand-collect administrative proceedings and litigation releases against all investment advisers between January 1, 2014, and December 31, 2021, from the SEC’s Web site (<https://www.sec.gov/enforcement/how-investigations-work>). After merging these two types of enforcement actions with the hedge fund advisers in our sample, we identified only two advisers from our sample with enforcement actions and neither involved disclosure-related issues.

To investigate whether the SEC pursues Form-ADV–related disclosure issues more generally, we extend our sample to examine enforcement actions against all investment advisers (not just hedge funds). Between 2014 and 2021, the number of brochure-related enforcement cases is relatively low for this population as well, consistent with the SEC not always reviewing the entire form. Specifically, we find that 3.6% of the total 1,770 administrative proceedings and litigation releases against investment advisers are caused by or related to adviser brochures.

Three reasons likely explain why regulatory enforcement actions involving hedge fund disclosure violations are rare. First, regulators have not historically allocated substantial resources to hedge fund supervision. This is reflected in the SEC’s decision to not formally review adviser brochures. Second, a prevalent view is that hedge fund investors are highly sophisticated and should have a solid understanding of the risks associated with these investments, which creates legal challenges to pursue such cases in

court. Third, it is difficult for investors, researchers, and even regulators to establish that the omitted information is material, as the information needed to evaluate the materiality of information omitted from Part 2 is not fully available to outsiders.

6. Additional Analyses

6.1 ALTERNATIVE EXPLANATIONS

The negative association between inconsistencies and fund performance is consistent with the hypothesis that advisers omit negative information from their brochures. In this section, we investigate several alternative explanations for this negative association, including that advisers are sloppy (i.e., either negligent or lack attention to detail); that advisers are overconfident; and that inconsistencies merely reflect past poor performance. Overall, the empirical evidence is consistent with at least some inconsistencies reflecting strategic omissions. We do not find strong support for overconfidence or sloppiness and the results cannot be solely explained by past fund performance.

6.1.1. Sloppiness. Inconsistent disclosures might arise due to sloppiness rather than adviser discretion. Funds characterized by such sloppiness could perform poorly if the sloppiness is indicative of poorer fund governance. We investigate this possibility in three tests.

6.1.1.1 Negative Information. We expect that advisers have greater incentives to strategically de-emphasize their most material issues. In contrast, if sloppiness explains the negative association between inconsistencies and fund performance, then the strength of that association should not vary with the materiality of the negative information. We separately analyze associations between fund performance and each type of inconsistency as well as subsamples that more likely contain material omissions. Negative performance for inconsistent funds with the most material omissions would be consistent with strategic nondisclosure.

In table 6, columns 1 and 2, we find a significantly negative association between fund returns and both *Inconsistent:Risk* and *Inconsistent:COI*, suggesting that both plausibly reflect omissions of material information. In column 3, we find that *Inconsistent:History* exhibits an insignificant association with fund returns. Examining this result further, we separate *Inconsistent:History* into different groups of varying materiality using two methods. First, we proxy for materiality using advisers who check multiple sections in Part 1 Item 11 and yet indicate they have no disciplinary history in Part 2 (*Inconsistent DH:2+Sections*). We find in column 1 of table IA.13 that such advisers have significantly worse performing funds, whereas funds of advisers who only check boxes in one section and have an inconsistency (*Inconsistent DH:1Section*) have performance similar to consistent funds.

Second, we consider a disciplinary event to be material if the event involves the advisory firm/management and the associated monetary

TABLE 6
Fund Performance by Inconsistency Type

	Alpha (Fund and Hsieh)							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Problem adviser	0.02 (0.13)	−0.07 (0.13)	−0.13 (0.13)	−0.15 (0.12)	−0.14 (0.12)	−0.15 (0.12)	−0.14 (0.12)	0.16 (0.14)
Inconsistent:Risk	−0.53*** (0.15)							−0.57*** (0.15)
Inconsistent:COI		−0.28** (0.14)						−0.33* (0.18)
Inconsistent:History			−0.14 (0.12)					−0.22* (0.13)
Inconsistent:Business				−0.01 (0.17)				−0.05 (0.16)
Inconsistent:Other					0.02 (0.12)			0.04 (0.12)
Inconsistent:Affiliations						−0.07 (0.11)		−0.02 (0.10)
Inconsistent:Custody							0.08 (0.09)	0.14 (0.10)
Observations	15,825	15,825	15,825	15,825	15,825	15,825	15,825	15,825
Adj R-squared	0.12	0.12	0.12	0.12	0.12	0.12	0.12	0.12
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Style × month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

This table analyzes the relation between hedge fund performance and disclosure inconsistency. We regress the Fung and Hsieh ([2001, 2004]) seven-factor alpha on an indicator for different types of inconsistent disclosures. We include as controls *Problem adviser*, *Log size*, and *Large family* as well as fund-style × month fixed effects. Standard errors (clustered by adviser) are in parentheses, and *, **, and *** indicate significance at the 0.1, 0.05, and 0.01 level, respectively.

$$\text{Alpha}_{i,t,m} = \alpha + \beta_1 \cdot \text{Inconsistency}_{i,t} + \gamma \cdot \text{Controls}_{i,t,m} + \phi_{\text{Style} \times \text{Month}} + \epsilon_{i,t,m}.$$

penalty is greater than or equal to the median penalty (*Inconsistent DH:Adviser&large penalty*).²⁵ In table IA.13, column 2, we find that such advisers have significantly worse performing funds, whereas funds with inconsistencies involving disciplinary histories with smaller penalties or an advisory affiliate (*Inconsistent DH:Affiliate or small penalty*) have performance similar to consistent funds.

6.1.1.2 Other Inconsistencies. We also expect that when information with neutral or positive performance implications is omitted from Part 2, it is more likely due to sloppiness rather than a strategic action. This is because there is no strong incentive for advisers to omit such information. A negative association between these inconsistencies and fund performance

²⁵ The median of the monetary penalty is \$40,000 for all disciplinary events and \$65,000 for disciplinary events related to an advisory firm or its management. We use \$65,000 for the empirical test and in untabulated analysis find that using the alternative benchmark gives us qualitatively similar results.

would suggest that sloppiness has consequences for fund performance (e.g., indicative of poor fund governance). Such an association would raise the concern that the negative association documented in table 3 involving the omissions of disciplinary histories, COIs, and risk discussions could also be due to sloppiness. In other words, if both types of inconsistencies are associated with fund performance, then sloppiness is a more viable explanation. On the other hand, the absence of an association between these truly sloppy inconsistencies and fund performance would suggest that the negative association documented in table 3 is not due to sloppiness. Absent other alternative explanations, such a finding would offer support for the interpretation that funds use discretion to specifically omit information with negative performance implications.

We first examine correlations between *Inconsistent: Any* and inconsistencies more likely due to sloppiness. As summarized in table IA.1, advisers are required to describe their business activities, affiliations, and custodial practices in both Part 1 and Part 2. This information is mostly about the basics of hedge fund advisers' business practices and likely does not have clear performance implications. Thus, we argue that any omissions of this information from Part 2 are more likely due to sloppiness than strategic discretion.²⁶

For each question in Part 1 Items 5, 6, 7, and 9, we evaluate whether hedge funds with an affirmative answer omitted this information from their Part 2 brochure. In untabulated results, we find that the disclosures of advisory business, other business activities, affiliations, and custody in Part 2 are sometimes inconsistent with those provided in Part 1. For example, 87.5% of advisers provide information in their brochure relating to their advisory business that has at least one inconsistency when compared with their Part 1 Item 5 disclosure (*Inconsistent: Business*). Inconsistency rates for Item 6 (*Inconsistent: Other*), Item 7 (*Inconsistent: Affiliations*), and Item 9 (*Inconsistent: Custody*) are 29.7%, 45.3%, and 24.8%, respectively. These four inconsistencies are only modestly correlated with *Inconsistent: Any* (results untabulated).

In contrast with the strategic inconsistencies examined in table 6, columns 1 to 3, columns 4 to 7 show that inconsistencies likely due to sloppiness are not associated with fund performance. This inference holds

²⁶ We examine the following items. Part 1 Item 5(B) requires advisers to report the number of employees that are registered representatives of a broker-dealer, are licensed agents of an insurance company or agency, or solicit advisory clients on behalf of the adviser. Item 5(D) asks advisers to provide information about their clients. Items 5E, 5F, 5G, 5I, and 5J relate to the types of services provided and fees charged to clients. Item 6 asks advisers to describe other business activities. Item 7 reports the financial industry affiliates of the advisers. Item 9 asks advisers to disclose their custodial practices. While some questions in Items 5, 6, and 7, such as those related to the affiliation with broker-dealers, could potentially have implications for COIs, hedge funds are required to disclose whether such relationships result in COIs in Item 8. Therefore, we do not consider these questions as specifically COI-related and do not separate them from other questions regarding advisers' business activities and affiliations.

when we include all inconsistency measures in column 8: *Inconsistent:Risk*, *Inconsistent:COI*, and *Inconsistent:History*, which plausibly reflect omissions of negative information, are each negatively associated with performance. In contrast, inconsistencies more plausibly due to sloppiness are not significantly associated with fund performance. These results support our hypothesis that advisers use discretion to de-emphasize negative information.

6.1.1.3 Brochure Textual Characteristics. We next analyze textual characteristics of adviser brochures to provide additional evidence that adviser discretion, rather than sloppiness, drives the main result. Prior studies suggest that managers can mislead investors and misrepresent investment prospects by decreasing emphasis on negative aspects of performance (Huang, Teoh, and Zhang [2014], Loughran and McDonald [2016]). If inconsistencies represent strategic actions by managers to de-emphasize problem areas, then we expect inconsistent advisers' brochures to have a decreased use of negative words. In contrast, there is little reason to expect that advisers systematically use fewer negative words if inconsistencies arise solely due to sloppiness.

Due to a high similarity of adviser brochures across years, we conduct this analysis at the adviser level.²⁷ We thus test whether the use of negative words differs for inconsistent funds using equation (3) (i.e., we include each fund once and use *Inconsistent:Any-max* to flag inconsistent funds).

In table 7, we find that inconsistent brochures have a smaller percentage of words that are negative, uncertain, constraining, litigious, modal weak, and modal moderate. Inconsistent brochures do not differ in their use of positive and modal strong words. We test whether the sum of *Inconsistent:Any-max* and *Problem adviser-max* significantly differs from zero and find that inconsistent brochures have a smaller percentage of negative and modal moderate words than non-problem advisers. Overall, the pattern is consistent with attempts by advisers with inconsistent brochures to de-emphasize fund problems.

6.1.2. Overconfidence. Prior research suggests that more experience and education are associated with greater perceived controllability and thus optimistic bias (Schrand and Zechman [2012]). In table IA.14, we analyze fund manager and adviser CCO characteristics, collected from LinkedIn. We find that fund managers of inconsistent funds do not differ in their fund experience, likelihood of being the CEO, or gender but have marginally less industry experience. CCOs of inconsistent funds are less likely to have a postgraduate degree (i.e., highest degree is bachelor's), which, according to prior research, should indicate lower overconfidence, but do not differ

²⁷We note that advisers' 2015 brochures are on average 94.6% similar to their 2014 brochures, based on identical text strings that are at least six words long and in both documents (calculated using Copyfind, see Cazier, McMullin, and Treu [2021]). Advisers' 2016, 2017, and 2018 brochures are on average 90.1%, 86.9%, and 82.9% similar to their 2014 brochures.

TABLE 7
Brochure Textual Characteristics

	Percent							
	(1) Neg	(2) Pos	(3) Uncertain	(4) Constrain	(5) Litigious	(6) Weak	(7) Mod	(8) Strong
Inconsistent:Any-max	-0.41*** (0.09)	-0.04 (0.05)	-0.30*** (0.07)	-0.10*** (0.03)	-0.13*** (0.04)	-0.08*** (0.02)	-0.09*** (0.03)	-0.03 (0.04)
Problem adviser-max	0.17 (0.12)	-0.01 (0.07)	0.16 (0.10)	0.14*** (0.04)	0.06 (0.05)	0.05** (0.02)	0.03 (0.03)	0.04 (0.05)
Avg log size	0.06*** (0.02)	-0.01 (0.01)	0.02 (0.02)	0.01 (0.01)	0.03*** (0.01)	0.00 (0.00)	0.01 (0.01)	-0.03*** (0.01)
Large family	0.13 (0.29)	-0.03 (0.11)	0.23 (0.16)	0.05 (0.10)	0.07 (0.12)	0.06 (0.04)	0.02 (0.07)	0.01 (0.06)
Observations	169	169	169	169	169	169	169	169
Adj <i>R</i> squared	0.27	-0.05	0.18	0.12	0.18	0.13	0.03	0.05
Style fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Problem adviser + Inconsistent = 0 (<i>p</i> -value)	0.03	0.34	0.20	0.37	0.12	0.12	0.07	0.86

This table analyzes textual attributes of Form ADV Part 2 (adviser brochures). The dependent variable in columns 1–8 is, respectively, the average percentage of an adviser's total brochure words that are negative, positive, uncertain, constraining, litigious, modal weak, modal moderate, and modal strong across all sample years. We include as controls *Inconsistent:Any-max*, an indicator if fund *i* has an adviser with an inconsistent disclosure in any sample year; *Problem adviser-max*, an indicator if fund *i* has an adviser with a problem in any sample year; *Avg log size*, the natural log of the average fund AUM across all sample years; and *Large family*, an indicator if the fund belongs to a fund family with more than 10 funds. We also include fund-style fixed effects. We tabulate in the table notes the *p*-value for an *F*-test that the sum of the coefficients on *Inconsistent:Any-max* and *Problem adviser-max* differs from zero. Robust standard errors are in parentheses, and *, **, ***, indicate statistical significance at the 0.1, 0.05, 0.01 level, respectively.

$$\text{Textual characteristic}_i = \alpha + \beta_1 \cdot \text{Inconsistent:Any-max}_i + \gamma \cdot \text{Controls}_i + \phi_{\text{style}} + \epsilon_i.$$

from CCOs of consistent funds in other characteristics. Overall, inconsistencies do not vary with most manager characteristics. The lack of a significant positive association with these measures of managerial overconfidence is inconsistent with overconfidence being the primary driver of inconsistent funds' lower performance.

6.1.3. Future Fund Performance. Another plausible explanation for a negative association between inconsistencies and concurrent performance is that inconsistencies reflect past poor performance. For example, advisers with poor-performing funds may be less inclined/able to retain a competent compliance officer than advisers with well-performing funds, which, in turn, may explain why the former's brochures are more likely to contain inconsistencies. In this case, inconsistencies reflect underlying fund problems already reflected in past performance.

We first examine whether inconsistencies, measured in year $t - 1$, are associated with lower performance in year t , conditional on performance in year $t - 1$. Because we lose all performance outcomes in 2014 (no measure of inconsistency from the prior year), we expand our sample to include fund performance in 2019 to maximize the sample size. Using this sample, we estimate the following regression:

$$\begin{aligned} \text{Alpha}_{i,t,m} = & \beta \cdot \text{Inconsistent:Any}_{i,t-1} + \gamma_1 \cdot \text{Problem adviser}_{i,t-1} + \gamma_2 \cdot \text{Log size}_{i,t,m} \\ & + \gamma_3 \cdot \text{Large family}_i + \gamma_4 \cdot \text{Past performance}_{i,t,m} + \phi_{\text{Style} \times \text{Month}} + \epsilon_{i,t,m}, \quad (4) \end{aligned}$$

where *Past performance* _{i,t,m} is the average performance of fund i over the six months prior to month m of year t and the remaining variables are previously defined. To ensure that brochures (and thus inconsistencies) are publicly observable, we match inconsistencies in year $t - 1$ to monthly fund returns in quarters 2–4 of year t and quarter 1 of year $t + 1$ to capture the extent to which inconsistencies in the previous year provide valuable signals of future performance.

Table 8 tabulates estimates of equation (4) for all four measures of fund performance. In all even-numbered columns, we include specifications that focus exclusively on funds with positive past performance (*Past performance* > 0) to isolate whether inconsistencies are associated with poorer future performance even when historical performance has been positive.²⁸ We find in all specifications that inconsistent funds have significantly lower future performance than consistent funds. Tests on the sum of *Inconsistent:Any* and *Problem adviser* indicate that inconsistent funds generally have significantly lower future performance than non–problem funds (see table notes for p -values). The coefficient on *Past performance* is also insignificant, suggesting that past performance is largely uncorrelated with future performance after accounting for disclosure inconsistency.

²⁸ Because the number of observations where performance is positive differs based on the performance measure used, the number of observations differs in each even-numbered column.

TABLE 8
Predicting Fund Performance

	Excess Return		Alpha (CAPM)		Alpha (Carhart)		Alpha (Fung and Hsieh)	
	(1) All	(2) Past perf > 0	(3) All	(4) Past perf > 0	(5) All	(6) Past perf > 0	(7) All	(8) Past perf > 0
Inconsistent:Any	-0.16* (0.09)	-0.25** (0.09)	-0.28** (0.10)	-0.39** (0.12)	-0.26** (0.10)	-0.38** (0.13)	-0.44** (0.12)	-0.57** (0.15)
Problem adviser	0.04 (0.09)	0.17* (0.09)	0.02 (0.10)	0.22* (0.12)	0.02 (0.11)	0.14 (0.12)	0.08 (0.12)	0.29** (0.14)
Log size	0.02 (0.01)	0.03** (0.02)	0.03** (0.02)	0.05** (0.02)	0.02 (0.02)	0.04* (0.02)	0.02 (0.02)	0.04* (0.02)
Large family	-0.06 (0.07)	0.00 (0.14)	0.09 (0.10)	0.07 (0.11)	0.08 (0.11)	0.16 (0.21)	0.00 (0.15)	-0.09 (0.16)
Past performance	0.02 (3.66)	7.06 (4.38)	3.39 (4.76)	0.66 (6.64)	2.97 (5.30)	0.73 (8.54)	-4.12 (4.22)	-4.38 (6.67)
Observations	17,436	10,877	17,436	9,377	17,436	8,946	17,436	9,125
Adj R-Squared	0.18	0.19	0.09	0.12	0.09	0.11	0.09	0.10
Style × month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Problem adviser + Inconsistent (p-value)	0.11	0.28	0.00	0.06	0.00	0.04	0.00	0.02

This table analyzes the ability of disclosure inconsistency to predict future fund performance. The sample is 392 funds with return and control-level data for the 2015–2019 sample period. Thus, relative to tables 2 and 3, we lose observations from 2014 (as we require a measure of inconsistency in the prior year) and include fund returns from 2019 (which are mapped to inconsistencies in 2018). The dependent variable is one of four performance measures for fund i in month m of year t , multiplied by 100 (see column headers). The explanatory variable of interest is *InconsistentAny* $_{i,t-1}$, an indicator variable equal to one if the fund had an inconsistent disclosure in the previous year. We add a quarter lag so that a fund's year t brochure (which typically is released during the first quarter of year $t + 1$) is available to investors (i.e., inconsistencies in year $t - 1$ are mapped to monthly returns from Q2–Q4 of year t and Q1 of year $t + 1$). Control variables include *Problem adviser*, *Log size*, *Large family*, *Past performance* (mean of the dependent variable over the previous six months), and fund-style × month fixed effects. Odd-numbered columns include all observations, and even-numbered columns include only fund months where the fund's past performance is positive (i.e., *Past performance* > 0); differences in the number of observations in all even-numbered columns reflects different subsets of funds that have positive performance per the specific performance measure. Standard errors (clustered by adviser) are in parentheses, and *, **, and *** indicate significance at the 0.1, 0.05, and 0.01 level, respectively.

$$\text{Alpha}_{i,t,m} = \alpha + \beta_1 \cdot \text{InconsistentAny}_{i,t-1} + \gamma \cdot \text{Controls}_{i,t,m} + \phi_{\text{Style} \times \text{Month}} + \epsilon_{i,t,m}.$$

We next analyze whether inconsistent funds are more likely to have lower future performance rankings. We use funds' future realized style-adjusted fractional rankings over the subsequent four quarters. In contrast to the analysis of monthly alphas, this specification allows us to identify funds with lower performance over longer horizons and examine whether inconsistencies reflect material fund or adviser problems. Specifically, we estimate the following quarterly level regression:

$$\begin{aligned} \text{Low FRANK}_{i,t,q+4} = & \alpha + \beta_1 \cdot \text{Inconsistent:Any}_{i,t-1} + \beta_2 \cdot \text{Problem adviser}_{i,t-1} \\ & + \beta_3 \cdot \text{FRANK decile}_{i,t,q} + \gamma \cdot \text{Controls}_{i,t,q} + \phi_{\text{Style} \times \text{Quarter}} + \epsilon_{i,t,q}. \end{aligned} \quad (5)$$

We use two specifications to capture lower future performance: an indicator if in any subsequent quarter $q+1$ through $q+4$ fund i will have a fractional performance ranking (*FRANK*) in the bottom decile; or an indicator if the fund will have a *FRANK* in the bottom decile specifically in quarter $q+4$.²⁹ By using inconsistencies from the prior year (which are already observable) and controlling for a fund's current performance (*FRANK decile* _{i,t,q} , an indicator variable for the fund's current decile ranking ranging from 1 to 10), these tests allow us to test whether inconsistencies are associated with the left tail of future performance. We continue to include controls for fund size and fund family as well as style-quarter fixed effects and cluster standard errors by adviser.

We tabulate estimates of equation (5) in table 9. In column 1, we analyze all fund quarters and find that inconsistencies are associated with an increased likelihood that fund performance will fall into the bottom decile at some point over the following year. To investigate whether the predictability of inconsistencies are limited to funds with current poor performance, in columns 2–3, we analyze the subset of funds with progressively higher fractional rankings as of quarter q . In both specifications, we find that regardless of current performance, inconsistencies indicate an increased likelihood that the fund will be in the bottom decile at some point over the following year. In columns 4–6, we similarly find that inconsistencies are predictive of low rankings exactly four quarters ahead. For instance, in column 6, we find that out of all funds with a current ranking greater than the 20th percentile, inconsistent funds are 66% more likely to be in the bottom decile in four quarters, calculated as the coefficient estimate of 0.06 divided by the average likelihood of 0.09 that a fund is in the bottom decile in four quarters.

We next examine whether inconsistent funds have lower survival rates. This test also provides an opportunity to further evaluate whether inconsistent funds cause investors financial harm. We first augment our sample to include all funds that deregister between 2014 and 2018. We then follow Brown et al. [2012b] and use TASS to identify funds with a defunct code as of 2018, where the fund has either closed down or TASS cannot determine

²⁹ The results are robust to including funds that deregister during the sample period.

TABLE 9
Low Fund Performance Rankings

	Q1 – Q4			Q4		
	(1)	(2)	(3)	(4)	(5)	(6)
	All	> Decile 1	> Decile 2	All	> Decile 1	> Decile 2
Inconsistent:Any	0.10** (0.04)	0.10** (0.04)	0.10** (0.04)	0.04* (0.02)	0.05*** (0.02)	0.06*** (0.02)
Problem adviser	0.09** (0.04)	0.08** (0.04)	0.09** (0.04)	0.03 (0.02)	0.02 (0.02)	0.02 (0.02)
FRANK decile	0.00 (0.00)	0.01** (0.00)	0.01** (0.01)	0.00 (0.00)	0.00 (0.00)	0.01* (0.00)
Log size	−0.03*** (0.01)	−0.03*** (0.01)	−0.02*** (0.01)	−0.01*** (0.00)	−0.01*** (0.00)	−0.01** (0.00)
Large family	−0.31*** (0.07)	−0.31*** (0.07)	−0.28*** (0.07)	−0.13*** (0.03)	−0.13*** (0.03)	−0.13*** (0.04)
Observations	4,648	4,183	3,716	3,150	2,836	2,527
Adj <i>R</i> -squared	0.13	0.14	0.15	0.02	0.03	0.04
Style × quarter fixed effects	Yes	Yes	Yes	Yes	Yes	Yes

This table analyzes the relation between inconsistencies and future fund performance rankings. The dependent variable is an indicator if fund *i* will have a fractional ranking (FRANK) in the bottom decile in quarter *q* + 1, *q* + 2, *q* + 3, or *q* + 4 (columns 1–3) or in quarter *q* + 4 (columns 4–6). The explanatory variables of interest are *Inconsistent:Any*_{*i,t*−1}, an indicator if the fund’s adviser had an inconsistent brochure in the previous year; *Problem adviser*_{*i,t*−1}, an indicator if the fund’s adviser had a disciplinary history, unusual COI, or high risk fund in the previous year; and *FRANK decile*_{*q*}, the fund’s fractional ranking in quarter *q*. We add a quarter lag so that a fund’s year *t* brochure (which typically is released during the first quarter of year *t* + 1) is available to investors (i.e., inconsistencies in year *t* − 1 are mapped to fractional rankings from Q2–Q4 of year *t* and Q1 of year *t* + 1). We also include controls for fund size and family size as well as fund-style × quarter fixed effects. Columns 1 and 4 analyze all quarterly observations, and columns 2, 3, 5, and 6 only use observations where the fund’s quarter *q* FRANK decile is greater than the number listed in the column title. The sample is 327 funds with available data for the 2015–2019 sample period (using inconsistencies identified in the 2014–2018 sample period). Standard errors (clustered by adviser) are in parentheses, and *, **, and *** indicate significance at the 0.1, 0.05, and 0.01 level, respectively.

$$\text{Low FRANK}_{i,t,q+4} = \alpha + \beta_1 \cdot \text{Inconsistent:Any}_{i,t-1} + \gamma \cdot \text{Controls}_{i,t,q} + \phi_{\text{Style} \times \text{Quarter}} + \epsilon_{i,t,q}.$$

the reason for why the fund stopped reporting returns. Of the 183 funds that deregister between 2014 and 2018, 139 (76%) are “dead” as of 2018. Thus, 24% of the funds in this expanded sample fail during our sample period.

In table 10, we analyze fund survival rates following disclosure inconsistencies. We regress an indicator for whether, as of year *t*, the fund will survive the following *X* years, *X* ∈ (2, 3, 4) (*AliveXyear*_{*i,t*}), on *Inconsistent:Any*_{*i,t*}, *Problem adviser*_{*i,t*}, *Log size*, *Large family*, and the fund’s average excess return in year *t* as well as style-year fixed effects. We continue to cluster standard errors by adviser. In column 1, we analyze two-year survival rates, dropping all observations in 2018 (as we do not know if the fund survives two years), and in columns 2 and 3, we analyze three- and four-year survival rates (dropping all observations after 2017 and 2016 in the respective specifications).

We find some evidence in columns 2 and 3 that inconsistent funds are marginally less likely to survive three or four years relative to consistent funds (i.e., are more likely to be dead by the end of the third or fourth

TABLE 10
Fund Survival

	(1) Alive 2 year	(2) Alive 3 year	(3) Alive 4 year
Inconsistent:Any	−0.04 (0.03)	−0.08* (0.04)	−0.09* (0.05)
Problem adviser	0.13*** (0.05)	0.21*** (0.07)	0.23*** (0.08)
Log size	0.03*** (0.01)	0.04*** (0.01)	0.04*** (0.01)
Large family	0.10* (0.05)	0.14* (0.07)	0.17** (0.08)
Average alpha	0.57 (0.43)	0.27 (0.48)	−0.08 (0.56)
Observations	1,822	1,436	980
Adj R-squared	0.13	0.15	0.16
Style-year fixed effects	Yes	Yes	Yes

This table analyzes the relation between fund survival and *Inconsistent:Any*_{*i,t*}. The dependent variable, *AliveXyear*_{*i,t*}, is an indicator if fund *i* will not be dead in *X* years, starting from year *t*. We include as controls *Problem adviser*_{*i,t*}, *Avg log size*, *Large family*, *Average alpha*_{*i,t*}, and style-year fixed effects. The sample is 575 funds (including 183 that deregister) covering the 2014–2018 sample period. To ensure sufficient data are available to estimate survival rates for the respective dependent variables, columns 1, 2, and 3 drop all observations after 2017, 2016, and 2015, respectively. Standard errors (clustered by adviser) are in parentheses, and *, **, and *** indicate significance at the 0.1, 0.05, and 0.01 level, respectively.

$$\text{AliveXyear}_{i,t} = \alpha + \beta_1 \cdot \text{Inconsistent:Any}_{i,t} + \gamma \cdot \text{Controls}_{i,t} + \phi_{\text{Style} \times \text{Year}} + \epsilon_{i,t}.$$

year), with *p*-values of 0.07 and 0.08, respectively. In terms of economic magnitudes, we find that funds are 9.7% less likely to survive three years relative to consistent funds. We also analyze the cross-section of funds (i.e., one observation per fund) and find in table IA.15 that funds that have been inconsistent during the sample period are more likely to be “dead” by the end of the sample and more likely to have deregistered during the sample period. To the extent that there is any measurement error in the performance measures previously analyzed, the evidence of an increased likelihood of fund failure reinforces the interpretation that a subset of problem advisers with material problems use discretion to de-emphasize information with adverse performance implications.

6.2 ALTERNATIVE TYPES OF DISCLOSURE INCONSISTENCY

6.2.1. *Form U4*. Our analyses thus far examine the use of discretion to omit negative performance information from Form ADV Part 2 that is observable in Part 1 and from historical return data. Investment adviser *representatives* are required to file additional forms, one of which is Form U4.³⁰ This section evaluates inconsistencies between disclosures of disciplinary histories in Form ADV Part 2 and Form U4.

³⁰ According to the Financial Industry Regulatory Authority (FINRA), “representatives of broker-dealers, investment advisers or issuers of securities must be registered with the appropriate jurisdictions and/or self-regulatory organizations (SROs).”

We follow two steps to examine whether Form ADV Part 2 omits any disciplinary histories listed in Form U4. We first identify all relevant hedge fund manager Form U4s by retrieving from the SEC's IAPD Web site all current and former employees associated with each hedge fund adviser. This process identified 346 unique hedge fund managers affiliated with 43 distinct hedge fund advisers. Of these 346 fund managers, 19 have disciplinary histories per the IAPD Web site, including five regulatory actions, one civil action, three criminal actions, seven customer disputes, and three miscellaneous actions. The majority of these events (18) do not have to be disclosed in Form ADV, as they occurred either more than 10 years earlier or while the manager worked for a different adviser, or else involve a customer dispute, which does not need to be disclosed in Form ADV. We check and find that the one remaining disciplinary history (a civil action) is disclosed in Form ADV.

Second, we check for consistency between Form ADV Part 2 and Form U4 by identifying those managers listed in Form ADV Part 2 as *not* having a disciplinary history. We identify 288 such fund managers listed in Part 2B (brochure supplement) associated with 30 distinct hedge fund advisers.³¹ After searching by manager name on both Broker-Check and the SEC's IAPD Web site, we find no evidence of any disciplinary history for these fund managers that should have been disclosed in Form ADV. We thus conclude that Form ADV Part 2 does not omit disciplinary histories listed in Form U4.

Finally, we investigate whether Form U4 omits disciplinary histories listed in Form ADV. We identified eight fund managers with a disciplinary history per Part 2 Item 9. A manual search by manager name confirms that six of these individuals have a disciplinary history per the IAPD Web site. The remaining two managers are not registered investment adviser representatives and thus do not have a Form U4. Overall, we conclude that there are minimal inconsistencies between Form ADV and Form U4.

6.2.2. Inconsistencies in the Opposite Direction. Our evidence thus far only involves omissions in Form ADV Part 2. This section analyzes inconsistencies that result when advisers discuss a disciplinary history or uncommon COI in their brochure that is not disclosed in their Form ADV Part 1 or provide a lengthy discussion of risks despite having only funds with low historical volatility. We do not expect these alternative inconsistencies relate to poorer fund performance. Specifically, as Form ADV Part 1 is reviewed by the SEC, omitting material information in Part 1 could draw regulatory scrutiny, which can be costly for advisers. Advisers thus have incentives to reduce such mistakes, implying that omissions

³¹ According to the SEC, certain registered hedge fund advisers are not required to file Form ADV Part 2B (brochure supplement). <https://www.sec.gov/about/forms/formadv-part2.pdf>

in Part 1 are unlikely to have material implications. Lengthy risk discussions, despite low historical return volatility, can be viewed as an effort to be more transparent and it is unlikely that funds would provide lengthy discussion of risks only to draw investor attention to material fund problems.

We first document the prevalence of inconsistencies in the “opposite” direction. Analyzing all advisers’ brochures, we identify 89 advisers who have an inconsistency in the opposite direction (i.e., omitted information from Part 1) at least once during our sample period, in contrast to the 98 advisers that omitted information from their brochure at least once. The two types of inconsistencies do not perfectly overlap: 40 advisers have inconsistencies in both directions at least once and 107 advisers have only one type of inconsistencies, leaving 43 advisers with neither type of inconsistency. The two types of inconsistencies likely have different underlying causes.

We next analyze the relation between both types of inconsistencies and fund performance using a modified version of equation (1). We use the variable *Problem adviser:Opp* to capture funds of advisers that disclose a disciplinary history or uncommon COI in their brochure or have all funds with historically low return volatility, and *Inconsistent:Opp* to capture funds of advisers that have an inconsistency in the opposite direction. In table IA.16, we find in all odd-numbered columns that *Inconsistent:Opp* is insignificantly associated with fund returns. Furthermore, in all even-numbered columns, we include both *Inconsistent:Any* and *Problem adviser* and find that only *Inconsistent:Any* is associated with lower performance. The results suggest that inconsistencies in the opposite direction are not associated with material fund problems, consistent with our expectations.

6.3 FUND RETURN MISREPORTING

Prior research finds that hedge funds misreport or manipulate their returns, as evidenced by a discontinuity at zero in a fund’s returns (*Kink*), a spike in reported returns for the month of December (*December spike*), a low correlation between a fund’s returns and the returns of other funds with the same investment style (*Low index beta*), and a low power of the Fung and Hsieh [2001,2004] factors in explaining a fund’s returns (*Low max R^2*) (Bollen and Pool [2009], Agarwal, Daniel, and Naik [2011], Bollen and Pool [2012], Dimmock and Gerken [2015], Honigsberg [2019]).³² This section examines the extent to which return misreporting relates to disclosure inconsistency. On the one hand, as inconsistent funds perform worse than other funds, they likely have greater incentives to misreport returns so that their performance will resemble other funds. If this is the case, our empirical results, which compare the performance of inconsistent funds

³² The literature also finds that liquidity factors explain these return patterns (Cassar and Gerakos [2011], Jorion and Schwarz [2014a]), motivating our use of style fixed effects.

with other funds, would be understated, as the performance of inconsistent funds would be even worse. On the other hand, as shown earlier, disclosure inconsistency helps advisers de-emphasize underlying fund problems that have not been reflected in current or past performance data. Thus, inconsistent funds may include those funds that do not yet have incentives to misreport current performance.

To examine this question, we analyze the relation between disclosure inconsistency and return misreporting at the fund level (using an indicator *Inconsistent: Any-max* for whether the fund's adviser has ever had an inconsistent disclosure), because each of the return misreporting measures must be estimated over a sample window. In table IA.17, we find an insignificant association between disclosure inconsistency and each of these measures. Furthermore, funds with problems are no more or less likely to engage in return misreporting. The overall evidence suggests that return misreporting and disclosure inconsistency are distinct phenomena.

6.4 STATE-REGISTERED ADVISERS

Our main analyses focus on advisers registered with the SEC. Qualified advisers with AUM below \$100 million must register with their state securities regulator. To evaluate whether the use of discretion we document is specific to SEC-registered hedge fund advisers, we collect a sample of state-registered hedge fund advisers by merging the list of state-registered investment advisers from the SEC's IAPD Web site with the Lipper TASS database. This process yields a sample of 41 state-registered hedge fund advisers (managing 67 funds).

As shown in table IA.18, six (15%) state-registered hedge fund advisers have a disciplinary history, one of which (17%) has an inconsistent disclosure in their Part 2 filings. Similarly, ten (24%) state-registered hedge fund advisers have an uncommon COI, four (40%) of which have an inconsistent disclosure. The rate of inconsistency is lower than that of SEC-registered advisers, which is 41% for disciplinary histories and 73% for COIs.

Many advisers registered with a state regulator do not report returns to TASS. Only 16 funds (out of 67) have return data available in the TASS data set. For this reason, we could not conduct meaningful statistical analyses of the relation between inconsistencies and fund performance based on a sample of 16 funds. One plausible reason for the lack of data is that the state-registered funds are smaller, and smaller funds are less likely to report returns to TASS.

Despite the lack of evidence on performance, our evidence does indicate that inconsistencies are not unique to SEC-registered advisers. Although it is uncertain whether the relation between inconsistencies and lower performance for SEC-registered funds would generalize to state-registered funds, there are reasons to expect this relation could generalize. State-registered funds are likely less monitored due to their smaller size. In addition, our analyses above indicate that these funds are less likely to report returns, potentially making it difficult for investors to evaluate the existence of in-

consistencies (e.g., inconsistencies regarding investment risks) and the implications of inconsistencies for performance. Further, studies show that funds that stop reporting returns tend to perform worse than those that choose to report (Agarwal, Fos, and Jiang [2013a], Aiken, Clifford, and Ellis [2013]).

6.5 ADDITIONAL ROBUSTNESS TESTS

We conduct two final robustness tests. First, we investigate whether the lower performance for funds with inconsistent risk disclosures concentrates in funds with overall short risk section/minimal use of uncertainty words. In table IA.19, we analyze associations between monthly fund returns and minimal risk disclosures using *Low discussion*, an indicator variable equal to one if the adviser's risk disclosure is in the bottom tercile. In column 1, we find that a short risk discussion or low use of uncertainty words is associated with lower fund performance; however, when we also include *Inconsistent:Risk* in column 2, we find that this association is primarily attributable to inconsistencies.

Second, prior research finds that hedge funds tend to backfill their returns after they start to report to commercial databases and that backfilled returns are usually higher than nonbackfilled returns (Jorion and Schwarz [2014b,2019]). Following Sialm, Sun, and Zheng [2020], we exclude the first 18 months of observations after a fund appears in TASS. In our monthly fund performance sample, 26 monthly returns representing six funds are thus eliminated and our main results are qualitatively unchanged.

7. Conclusion

In an effort to increase transparency in the hedge fund industry, the Dodd-Frank Act of 2010 mandated that hedge fund advisers register with the SEC and provide disclosures of their business and operations, including material operational and investment risks. One of these disclosures, Form ADV Part 2 or the adviser brochure, is not formally reviewed by the SEC and does not explicitly define what constitutes material information. We document that approximately 40% of advisers have omissions of operational and investment risks in their brochures. Funds with such inconsistencies realize lower current and future fund performance and have a higher likelihood of fund failure, suggesting the omissions on average reflect strategic efforts to de-emphasize negative information.

Sophisticated market participants, including lenders, equity investors, reputable auditors, and clients do not seem to respond to the inconsistencies, despite these funds' poorer future performance and higher failure rates. The evidence is consistent with limited strategic thinking leading to underreactions to the implications of strategic actions. One plausible reason for limited strategic thinking is that SEC regulation requires material information to be disclosed in Part 2, which may lead market participants to

believe that the omitted information is immaterial. We also document low SEC enforcement risk against inconsistent disclosures, likely attributable to the lack of formal review of brochures, legal challenges pursuing cases against hedge fund advisers, and difficulty proving that inconsistencies reflect noncompliance. Collectively, due to limited strategic thinking and low enforcement risk, fund managers can successfully de-emphasize fund problems without sacrificing fund flows or deterring market participants.

We find that hedge fund investors do not react to the implications of inconsistencies, casting doubt on the commonly held view that they are sophisticated. This finding raises important questions regarding the foundational assumption that informed, capable investors can adequately assess and manage risk and, thus, that minimal regulatory oversight is needed. Nonetheless, we caution that advocating for stricter regulation would require a deeper understanding of the underlying causes of market failures, clarity on the appropriate regulatory scope (e.g., definitions of noncompliance and enforcement mechanisms), and carefully weighing regulatory costs against potential benefits.

APPENDIX A

Variable Definitions

Variable	Definition
<i>Performance and Flow Measures</i>	
Excess return	Fund monthly net-of-fee returns reported in Lipper TASS in excess of risk-free returns (in percentage points).
Alpha (CAPM)	Monthly alpha (in percentage points) from the CAPM model, estimated over a rolling window of 24 months and requiring at least 9 months. We regress fund monthly net-of-fee returns reported in Lipper TASS (in excess of risk-free returns) on CRSP's value-weighted returns. Betas are calculated using a prior 24 month window and alphas are calculated as current-month abnormal returns.
Alpha (Carhart)	Monthly alpha (in percentage points) from the Carhart [1997] four-factor model, estimated over a rolling window of 24 months and requiring at least 9 months. We regress fund returns on (1) excess return on the market, (2) small-minus-big (SMB) return, (3) high-minus-low (HML) return, and (4) a momentum factor (UMD) (in place of PRIYR in Carhart [1997] for the ease of construction). Betas are calculated using a prior 24 month window and alphas are calculated as current-month abnormal returns.

Variable	Definition
Alpha (Fung and Hsieh)	Monthly alpha (in percentage points) from the Fung and Hsieh [2001, 2004] seven-factor model, estimated over a rolling window of 24 months and requiring at least 9 months. We regress fund returns on (1) an equity market factor (the Standard & Poor's 500 index monthly total return less the risk-free rate), (2) a size spread factor (Russell 2000 index monthly total return less Standard & Poor's 500 monthly total return), (3) a bond market factor (the monthly change in the 10-year Treasury constant maturity yield), (4) a credit spread factor (the monthly change in the Moody's Baa yield less 10-year Treasury constant maturity yield), (5) a bond-trend following factor, (6) a currency-trend following factor, and (7) a commodity-trend following factor. The bond-trend following factor, currency-trend following factor, and commodity-trend following factor are updated monthly and available from David Hsieh's Data Library (https://faculty.fuqua.duke.edu/~dah7/HFRFData.htm). Betas are calculated using a prior 24 month window and alphas are calculated as current-month abnormal returns.
Style adjusted	Monthly fund return less the average return for funds with the same investment style (in percentage points).
Fund flow	Fund quarterly flows, computed as $[AUM(q) - AUM(q - 1) \times (Returns(q) + 1) / AUM(q - 1)]$. AUM (q) is the fund AUM at the end of quarter q. Returns (q) are calculated by compounding $(Returns(m) + 1)$ within quarter q.
FRANK	The fractional ranking of quarterly fund performance. Specifically, $FRANK1 = \min (FRANK, 1/3)$, $FRANK2 = \min (FRANK - FRANK1, 1/3)$, and $FRANK3 = \min (FRANK - FRANK1 - FRANK2, 1/3)$.
<i>Measures of Inconsistency</i>	
Problem adviser $_{i,t}$	Indicator variable that takes a value of one if fund i 's adviser has a disciplinary history or uncommon COI per its year t Form ADV Part 1 or if the standard deviation of monthly returns over the previous three years of at least one of the adviser's funds is in the top tercile.
Inconsistent:COI $_{i,t}$	Indicator variable that takes a value of one if fund i 's adviser has an uncommon external conflict of interest (COI) per its year t Form ADV Part 1 Item 8 and does not disclose the COI in its contemporaneous brochure.

Variable	Definition
Inconsistent:Risk _{<i>i,t</i>}	Indicator variable that takes a value of one if fund <i>i</i> 's adviser has a brochure in year <i>t</i> with a risk subsection or uncertainty-related words (scaled by the total number of words) in the respective bottom tercile and at least one of the adviser's funds has a standard deviation of monthly fund returns over the previous three years in the top tercile.
Inconsistent:History _{<i>i,t</i>}	Indicator variable that takes a value of one if fund <i>i</i> 's adviser checks any boxes in its year <i>t</i> Form ADV Part 1 Item 11 indicating past disciplinary event(s) and also discloses that it has no disciplinary history in its contemporaneous brochure.
Inconsistent:1Section _{<i>i,t</i>}	Indicator variable that takes a value of one if fund <i>i</i> 's adviser checks boxes in only one section of its year <i>t</i> Form ADV Part 1 Item 11 and also discloses that it has no disciplinary history in its contemporaneous brochure.
Inconsistent:2+Sections _{<i>i,t</i>}	Indicator variable that takes a value of one if fund <i>i</i> 's adviser checks boxes in two or more sections of its year <i>t</i> Form ADV Part 1 Item 11 and also discloses that it has no disciplinary history in its contemporaneous brochure.
Inconsistent:Adviser & large penalty _{<i>i,t</i>}	Indicator variable that takes a value of one if fund <i>i</i> 's adviser has a disciplinary history related to the advisory firm and/or management persons and has a monetary penalty greater than or equal to the sample median, and also discloses that it has no disciplinary history (or does not disclose the monetary penalty) in its contemporaneous brochure.
Inconsistent:Affiliate or small penalty _{<i>i,t</i>}	Indicator variable that takes a value of one if fund <i>i</i> 's adviser has a disciplinary history only related to the advisory affiliates or has a monetary penalty below the sample median, and also discloses that it has no disciplinary history in its contemporaneous brochure.
Inconsistent:Business _{<i>i,t</i>}	Indicator variable that takes a value of one if fund <i>i</i> 's adviser checks one or more boxes in its year <i>t</i> Form ADV Part 1 Item 5 (B2, B5, B6, D, E, F, G, I, J) and does not disclose related business information in its contemporaneous brochure.
Inconsistent:Other _{<i>i,t</i>}	Indicator variable that takes a value of one if fund <i>i</i> 's adviser checks one or more boxes in its year <i>t</i> Form ADV Part 1 Item 6 and does not disclose that it engages in other business activities in its contemporaneous brochure.

Variable	Definition
Inconsistent:Affiliations $_{i,t}$	Indicator variable that takes a value of one if fund i 's adviser checks one or more boxes in its year t Form ADV Part 1 Item 7 and does not disclose related financial industry affiliations or private fund information in its contemporaneous brochure.
Inconsistent:Custody $_{i,t}$	Indicator variable that takes a value of one if fund i 's adviser checks one or more boxes in its year t Form ADV Part 1 Item 9 and does not disclose related custody information in its contemporaneous brochure.
Inconsistent:Any $_{i,t}$	Indicator variable that takes a value of one if any of the three inconsistency indicators (Inconsistent:COI, Inconsistent:Risk, Inconsistent:History) for fund i in year t equals one.
Inconsistent:Any-max $_i$	Indicator variable that takes a value of one if fund i has any inconsistency in any of the sample years.
Problem adviser-max $_i$	Indicator variable that takes a value of one if fund i has an adviser that had in any of the sample years a disciplinary history, uncommon COI, or fund with a standard deviation of monthly returns over the previous three years in the top tercile.
<i>Fund Characteristics</i>	
Age	Number of years since inception of hedge fund.
Number of funds	Number of funds within a fund family.
Size	Fund's assets under management (AUM) in millions of dollars.
Large family	Indicator variable that takes a value of one if a fund belongs to a fund family with more than ten funds.
Management fees	Percentage of fund's net asset value charged by the hedge fund.
Incentive fees	Percentage of profits charged by the hedge fund.
Highwater mark	Indicator variable that takes a value of one if a fund uses a highwater mark to calculate incentive fees.
Lockup period	Number of days before an investment can be withdrawn.
Redemption notice period	Number of days required to inform a hedge fund of a future withdrawal.
Total restrictions	The sum of lockup period and redemption notice period, following prior literature (e.g., Patton, Ramadorai, and Streatfield [2015]).
Redemption frequency	Minimum number of days required between two withdrawals.
Subscription period	Number of times per year an investor can invest into a fund.
Minimum investment	Minimum amounts of investment required by the fund (in millions).
Payout period	Number of days between redemption and when investors get paid.
Open to public	Indicator variable that takes a value of one if a fund is open to public.

Variable	Definition
Foreign	Indicator variable that takes a value of one if a fund is located outside the US.
Fiduciary	Indicator variable that takes a value of one if an adviser mentions in the brochure that it also works as a trustee for investors.
Leveraged	Indicator variable that takes a value of one if a fund uses leverage.
Max lev	A fund's maximum leverage ratio.
Avg lev	A fund's average leverage ratio.
Margin	Indicator variable that takes a value of one if a fund uses leverage by borrowing on margin.
Big 4	Indicator variable that takes a value of one if a fund uses a Big 4 auditor.
Direct owners	Number of direct owners of an investment adviser.
Indirect owners	Number of indirect owners of an investment adviser.
Controlling	Number of controlling owners of an investment adviser.
75%Num	Number of direct owners who own at least 75% of the investment adviser.
75%Pct	Percentage of direct owners who own at least 75% of the investment adviser.
Domestic direct	Number of domestic corporations listed as direct owners.
InvestCo	Indicator variable equal to one if the investment adviser lists an investment company as a client per its Form ADV Part 1.
Pension	Indicator variable equal to one if the investment adviser lists a pension and profit-sharing plan as a client per its Form ADV Part 1.
Discontinuity at zero (Kink)	Indicator variable that takes a value of one if a fund has a kink at zero in the pooled distributions of its monthly returns. We create three bins surrounding zero (large losses, small losses, small gains) using bandwidths suggested by Bollen and Pool [2009]. A fund is considered to have a kink at zero if the number of small losses is significantly lower than the average of the two surrounding bins with a <i>t</i> -statistic of 1.65 or greater. We require a fund to have at least 36 months of reported performance data.
December spike	Indicator variable that takes a value of one if a fund's returns spike in the month of December. We regress each fund's monthly returns on the Fung and Hsieh [2001, 2004] seven risk factors, an indicator for the month of December, and year fixed effects. A fund is considered to have a December spike if the coefficient on the December indicator is significantly positive with a <i>t</i> -statistic of 1.65 or greater. We require a fund to have at least 36-months of reported performance data.

Variable	Definition
Low index beta	Indicator variable that takes a value of one if a fund's monthly returns have low correlation with the returns of other funds with the same investment style. Following Bollen and Pool [2012] and Dimmock and Gerken [2015], we regress a fund's monthly returns on the average monthly returns of other funds that have the same investment style in our sample. This misreporting flag is triggered if the regression coefficient has a <i>t</i> -statistic less than 1.65.
Low max R^2	Indicator variable that takes a value of one if the Fung and Hsieh [2001, 2004] factors have lower explanatory power for a fund's actual monthly returns than for the simulated fund returns. Following Bollen and Pool [2012] and Dimmock and Gerken [2015], we obtain the maximum adjusted R -squared by regressing fund excess returns on all possible sets of three out of seven Fung and Hsieh [2001, 2004] factors. We repeat this process 100 times using simulated returns with mean and standard deviation equal to the realized mean and standard deviation. The misreporting flag is triggered if the maximum adjusted R -squared based on the actual fund returns is below the 90th percentile of the maximum adjusted R -squared based on the simulated returns.
Fund exp	Number of years the fund manager or the CCO has worked for the fund as of the end of 2018.
Ind exp	Number of years the fund manager or the CCO has worked for the industry as of the end of 2018.
CEO	Indicator variable that takes a value of one if the fund manager or the CCO is also the CEO, president, founder, or principal of the fund.
Masters degree	Indicator variable that takes a value of one if the fund manager or the CCO holds a master's degree or above.
Male	Indicator variable that takes a value of one if the fund manager or the CCO is male.
Neg	Percentage of total brochure words that are negative per the Loughran and McDonald word lists (hereinafter).
Pos	Percentage of total brochure words that are positive.
Uncertain	Percentage of total brochure words that are uncertain.
Constrain	Percentage of total brochure words that are constraining.
Litigious	Percentage of total brochure words that are litigious.
Weak	Percentage of total brochure words that are modal weak.

Variable	Definition
Mod	Percentage of total brochure words that are modal moderate.
Strong	Percentage of total brochure words that are modal strong.
Alive X year $_{i,t}$	Indicator if fund i will not be dead in X years, starting with year t .
Inconsistent:Opp $_{i,t}$	Indicator if the fund discusses a disciplinary history or uncommon COI in its year t brochure that is not listed in its Form ADV Part 1 or has a risk discussion or uncertainty words in the top tercile and a historical return volatility in the bottom tercile.
Problem Adviser:Opp $_{i,t}$	Indicator if the fund discusses a disciplinary history or uncommon COI in its brochure or has a historical return volatility in the bottom tercile.
Low FRANK $_{i,t,q+4}$	Indicator if fund i will have a fractional ranking (FRANK) in the bottom decile in quarter $q + 1$, $q + 2$, $q + 3$, or $q + 4$ of year t .

This table provides the definition and calculation of variables used in the analyses.

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